

Project 2 Data 624

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Project 2

Importing the data

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.2
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
## Warning: package 'tibble' was built under R version 4.5.2
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.6
```

```
## v forcats   1.0.1      v stringr   1.5.2
```

```
## v ggplot2   4.0.1      v tibble    3.3.1
```

```
## v lubridate 1.9.4      v tidyr     1.3.1
```

```
## v purrr     1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(VIM)
```

```
## Warning: package 'VIM' was built under R version 4.5.3
```

```
## Loading required package: colorspace
```

```
## Warning: package 'colorspace' was built under R version 4.5.3
```

```
## Loading required package: grid
```

```
## VIM is ready to use.
```

```
##
```

```
## Suggestions and bug-reports can be submitted at: https://github.com/statistikat/VIM/issues
```

```
##
## Attaching package: 'VIM'
##
## The following object is masked from 'package:datasets':
##
##     sleep
```

```
test_data <- read.csv("https://raw.githubusercontent.com/samanthabarbaro/data624/refs/heads/main/testdata.csv")
training_data <- read.csv("https://raw.githubusercontent.com/samanthabarbaro/data624/refs/heads/main/trainingdata.csv")
```

```
glimpse(test_data)
```

```
## Rows: 267
## Columns: 33
## $ Brand.Code      <chr> "D", "A", "B", "B", "B", "B", "A", "B", "A", "D", "B~
## $ Carb.Volume     <dbl> 5.480000, 5.393333, 5.293333, 5.266667, 5.406667, 5.~
## $ Fill.Ounces     <dbl> 24.03333, 23.95333, 23.92000, 23.94000, 24.20000, 24~
## $ PC.Volume       <dbl> 0.2700000, 0.2266667, 0.3033333, 0.1860000, 0.160000~
## $ Carb.Pressure   <dbl> 65.4, 63.2, 66.4, 64.8, 69.4, 73.4, 65.2, 67.4, 66.8~
## $ Carb.Temp       <dbl> 134.6, 135.0, 140.4, 139.0, 142.2, 147.2, 134.6, 139~
## $ PSC             <dbl> 0.236, 0.042, 0.068, 0.004, 0.040, 0.078, 0.088, 0.0~
## $ PSC.Fill       <dbl> 0.40, 0.22, 0.10, 0.20, 0.30, 0.22, 0.14, 0.10, 0.48~
## $ PSC.CO2        <dbl> 0.04, 0.08, 0.02, 0.02, 0.06, NA, 0.00, 0.04, 0.04, ~
## $ Mnf.Flow       <dbl> -100, -100, -100, -100, -100, -100, -100, -100, -100~
## $ Carb.Pressure1  <dbl> 116.6, 118.8, 120.2, 124.8, 115.0, 118.6, 117.6, 121~
## $ Fill.Pressure   <dbl> 46.0, 46.2, 45.8, 40.0, 51.4, 46.4, 46.2, 40.0, 43.8~
## $ Hyd.Pressure1   <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Hyd.Pressure2   <dbl> NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ Hyd.Pressure3   <dbl> NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ Hyd.Pressure4   <int> 96, 112, 98, 132, 94, 94, 108, 108, 110, 106, 98, 96~
## $ Filler.Level    <dbl> 129.4, 120.0, 119.4, 120.2, 116.0, 120.4, 119.6, 131~
## $ Filler.Speed    <int> 3986, 4012, 4010, NA, 4018, 4010, 4010, NA, 4010, 10~
## $ Temperature    <dbl> 66.0, 65.6, 65.6, 74.4, 66.4, 66.6, 66.8, NA, 65.8, ~
## $ Usage.cont      <dbl> 21.66, 17.60, 24.18, 18.12, 21.32, 18.00, 17.68, 12.~
## $ Carb.Flow       <int> 2950, 2916, 3056, 28, 3214, 3064, 3042, 1972, 2502, ~
## $ Density         <dbl> 0.88, 1.50, 0.90, 0.74, 0.88, 0.84, 1.48, 1.60, 1.52~
## $ MFR             <dbl> 727.6, 735.8, 734.8, NA, 752.0, 732.0, 729.8, NA, 74~
## $ Balling         <dbl> 1.398, 2.942, 1.448, 1.056, 1.398, 1.298, 2.894, 3.3~
## $ Pressure.Vacuum <dbl> -3.8, -4.4, -4.2, -4.0, -4.0, -3.8, -4.2, -4.4, -4.4~
## $ PH              <lg1> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ Oxygen.Filler   <dbl> 0.022, 0.030, 0.046, NA, 0.082, 0.064, 0.042, 0.096,~
## $ Bowl.Setpoint   <int> 130, 120, 120, 120, 120, 120, 120, 120, 120, 120, 12~
## $ Pressure.Setpoint <dbl> 45.2, 46.0, 46.0, 46.0, 50.0, 46.0, 46.0, 46.0, 46.0~
## $ Air.Pressurer    <dbl> 142.6, 147.2, 146.6, 146.4, 145.8, 146.0, 145.0, 146~
## $ Alch.Rel        <dbl> 6.56, 7.14, 6.52, 6.48, 6.50, 6.50, 7.18, 7.16, 7.14~
## $ Carb.Rel        <dbl> 5.34, 5.58, 5.34, 5.50, 5.38, 5.42, 5.46, 5.42, 5.44~
## $ Balling.Lvl     <dbl> 1.48, 3.04, 1.46, 1.48, 1.46, 1.44, 3.02, 3.00, 3.10~
```

```
glimpse(training_data)
```

```
## Rows: 2,571
```

```
## Columns: 33
## $ Brand.Code      <chr> "B", "A", "B", "A", "A", "A", "A", "B", "B", "B", "B~
## $ Carb.Volume     <dbl> 5.340000, 5.426667, 5.286667, 5.440000, 5.486667, 5.~
## $ Fill.Ounces     <dbl> 23.96667, 24.00667, 24.06000, 24.00667, 24.31333, 23~
## $ PC.Volume       <dbl> 0.2633333, 0.2386667, 0.2633333, 0.2933333, 0.111333~
## $ Carb.Pressure   <dbl> 68.2, 68.4, 70.8, 63.0, 67.2, 66.6, 64.2, 67.6, 64.2~
## $ Carb.Temp       <dbl> 141.2, 139.6, 144.8, 132.6, 136.8, 138.4, 136.8, 141~
## $ PSC             <dbl> 0.104, 0.124, 0.090, NA, 0.026, 0.090, 0.128, 0.154,~
## $ PSC.Fill       <dbl> 0.26, 0.22, 0.34, 0.42, 0.16, 0.24, 0.40, 0.34, 0.12~
## $ PSC.CO2        <dbl> 0.04, 0.04, 0.16, 0.04, 0.12, 0.04, 0.04, 0.04, 0.14~
## $ Mnf.Flow       <dbl> -100, -100, -100, -100, -100, -100, -100, -100, -100~
## $ Carb.Pressure1  <dbl> 118.8, 121.6, 120.2, 115.2, 118.4, 119.6, 122.2, 124~
## $ Fill.Pressure   <dbl> 46.0, 46.0, 46.0, 46.4, 45.8, 45.6, 51.8, 46.8, 46.0~
## $ Hyd.Pressure1   <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Hyd.Pressure2   <dbl> NA, NA, NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Hyd.Pressure3   <dbl> NA, NA, NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Hyd.Pressure4   <int> 118, 106, 82, 92, 92, 116, 124, 132, 90, 108, 94, 86~
## $ Filler.Level    <dbl> 121.2, 118.6, 120.0, 117.8, 118.6, 120.2, 123.4, 118~
## $ Filler.Speed    <int> 4002, 3986, 4020, 4012, 4010, 4014, NA, 1004, 4014, ~
## $ Temperature     <dbl> 66.0, 67.6, 67.0, 65.6, 65.6, 66.2, 65.8, 65.2, 65.4~
## $ Usage.cont      <dbl> 16.18, 19.90, 17.76, 17.42, 17.68, 23.82, 20.74, 18.~
## $ Carb.Flow       <int> 2932, 3144, 2914, 3062, 3054, 2948, 30, 684, 2902, 3~
## $ Density         <dbl> 0.88, 0.92, 1.58, 1.54, 1.54, 1.52, 0.84, 0.84, 0.90~
## $ MFR             <dbl> 725.0, 726.8, 735.0, 730.6, 722.8, 738.8, NA, NA, 74~
## $ Balling         <dbl> 1.398, 1.498, 3.142, 3.042, 3.042, 2.992, 1.298, 1.2~
## $ Pressure.Vacuum <dbl> -4.0, -4.0, -3.8, -4.4, -4.4, -4.4, -4.4, -4.4, -4.4~
## $ PH              <dbl> 8.36, 8.26, 8.94, 8.24, 8.26, 8.32, 8.40, 8.38, 8.38~
## $ Oxygen.Filler   <dbl> 0.022, 0.026, 0.024, 0.030, 0.030, 0.024, 0.066, 0.0~
## $ Bowl.Setpoint   <int> 120, 120, 120, 120, 120, 120, 120, 120, 120, 120, 12~
## $ Pressure.Setpoint <dbl> 46.4, 46.8, 46.6, 46.0, 46.0, 46.0, 46.0, 46.0, 46.0~
## $ Air.Pressurer   <dbl> 142.6, 143.0, 142.0, 146.2, 146.2, 146.6, 146.2, 146~
## $ Alch.Rel        <dbl> 6.58, 6.56, 7.66, 7.14, 7.14, 7.16, 6.54, 6.52, 6.52~
## $ Carb.Rel        <dbl> 5.32, 5.30, 5.84, 5.42, 5.44, 5.44, 5.38, 5.34, 5.34~
## $ Balling.Lvl     <dbl> 1.48, 1.56, 3.28, 3.04, 3.04, 3.02, 1.44, 1.44, 1.44~
```

```
str(training_data)
```

```
## 'data.frame': 2571 obs. of 33 variables:
## $ Brand.Code : chr "B" "A" "B" "A" ...
## $ Carb.Volume : num 5.34 5.43 5.29 5.44 5.49 ...
## $ Fill.Ounces : num 24 24 24.1 24 24.3 ...
## $ PC.Volume : num 0.263 0.239 0.263 0.293 0.111 ...
## $ Carb.Pressure : num 68.2 68.4 70.8 63 67.2 66.6 64.2 67.6 64.2 72 ...
## $ Carb.Temp : num 141 140 145 133 137 ...
## $ PSC : num 0.104 0.124 0.09 NA 0.026 0.09 0.128 0.154 0.132 0.014 ...
## $ PSC.Fill : num 0.26 0.22 0.34 0.42 0.16 0.24 0.4 0.34 0.12 0.24 ...
## $ PSC.CO2 : num 0.04 0.04 0.16 0.04 0.12 0.04 0.04 0.04 0.14 0.06 ...
## $ Mnf.Flow : num -100 -100 -100 -100 -100 -100 -100 -100 -100 -100 ...
## $ Carb.Pressure1 : num 119 122 120 115 118 ...
## $ Fill.Pressure : num 46 46 46 46.4 45.8 45.6 51.8 46.8 46 45.2 ...
## $ Hyd.Pressure1 : num 0 0 0 0 0 0 0 0 0 0 ...
## $ Hyd.Pressure2 : num NA NA NA 0 0 0 0 0 0 0 ...
## $ Hyd.Pressure3 : num NA NA NA 0 0 0 0 0 0 0 ...
## $ Hyd.Pressure4 : int 118 106 82 92 92 116 124 132 90 108 ...
```

```
## $ Filler.Level      : num  121 119 120 118 119 ...
## $ Filler.Speed      : int   4002 3986 4020 4012 4010 4014 NA 1004 4014 4028 ...
## $ Temperature      : num   66 67.6 67 65.6 65.6 66.2 65.8 65.2 65.4 66.6 ...
## $ Usage.cont        : num   16.2 19.9 17.8 17.4 17.7 ...
## $ Carb.Flow         : int   2932 3144 2914 3062 3054 2948 30 684 2902 3038 ...
## $ Density           : num   0.88 0.92 1.58 1.54 1.54 1.52 0.84 0.84 0.9 0.9 ...
## $ MFR               : num   725 727 735 731 723 ...
## $ Balling           : num    1.4 1.5 3.14 3.04 3.04 ...
## $ Pressure.Vacuum   : num   -4 -4 -3.8 -4.4 -4.4 -4.4 -4.4 -4.4 -4.4 ...
## $ PH                : num    8.36 8.26 8.94 8.24 8.26 8.32 8.4 8.38 8.38 8.5 ...
## $ Oxygen.Filler     : num    0.022 0.026 0.024 0.03 0.03 0.024 0.066 0.046 0.064 0.022 ...
## $ Bowl.Setpoint     : int   120 120 120 120 120 120 120 120 120 120 ...
## $ Pressure.Setpoint : num   46.4 46.8 46.6 46 46 46 46 46 46 46 ...
## $ Air.Pressurer     : num   143 143 142 146 146 ...
## $ Alch.Rel          : num    6.58 6.56 7.66 7.14 7.14 7.16 6.54 6.52 6.52 6.54 ...
## $ Carb.Rel          : num    5.32 5.3 5.84 5.42 5.44 5.44 5.38 5.34 5.34 5.34 ...
## $ Balling.Lvl       : num    1.48 1.56 3.28 3.04 3.04 3.02 1.44 1.44 1.44 1.38 ...
```

All the rows with numbers are formatted as numbers. All the variables are numeric except for brand code, likely the only categorical variable.

Exploring the data

We'll look at:

- The number of cases
- Missing values
- Data types (numerical vs. categorical)

Number of cases

```
nrow(training_data)
```

```
## [1] 2571
```

```
nrow(test_data)
```

```
## [1] 267
```

The training data has 2571 cases and the test data has 267.

Variables

Counting the number of columns/variables:

```
ncol(test_data)
```

```
## [1] 33
```

```
ncol(training_data)
```

```
## [1] 33
```

Both data sets have 33 columns.

Are all the variables represented in both data sets?

```
colnames(test_data)
```

```
## [1] "Brand.Code"      "Carb.Volume"      "Fill.Ounces"
## [4] "PC.Volume"       "Carb.Pressure"    "Carb.Temp"
## [7] "PSC"             "PSC.Fill"         "PSC.CO2"
## [10] "Mnf.Flow"        "Carb.Pressure1"   "Fill.Pressure"
## [13] "Hyd.Pressure1"   "Hyd.Pressure2"    "Hyd.Pressure3"
## [16] "Hyd.Pressure4"   "Filler.Level"     "Filler.Speed"
## [19] "Temperature"     "Usage.cont"       "Carb.Flow"
## [22] "Density"         "MFR"              "Balling"
## [25] "Pressure.Vacuum" "PH"               "Oxygen.Filler"
## [28] "Bowl.Setpoint"   "Pressure.Setpoint" "Air.Pressurer"
## [31] "Alch.Rel"        "Carb.Rel"         "Balling.Lvl"
```

```
colnames(training_data)
```

```
## [1] "Brand.Code"      "Carb.Volume"      "Fill.Ounces"
## [4] "PC.Volume"       "Carb.Pressure"    "Carb.Temp"
## [7] "PSC"             "PSC.Fill"         "PSC.CO2"
## [10] "Mnf.Flow"        "Carb.Pressure1"   "Fill.Pressure"
## [13] "Hyd.Pressure1"   "Hyd.Pressure2"    "Hyd.Pressure3"
## [16] "Hyd.Pressure4"   "Filler.Level"     "Filler.Speed"
## [19] "Temperature"     "Usage.cont"       "Carb.Flow"
## [22] "Density"         "MFR"              "Balling"
## [25] "Pressure.Vacuum" "PH"               "Oxygen.Filler"
## [28] "Bowl.Setpoint"   "Pressure.Setpoint" "Air.Pressurer"
## [31] "Alch.Rel"        "Carb.Rel"         "Balling.Lvl"
```

Both data sets have the same columns, so we can proceed.

Missing data

How many missing values are there? Are there any trends? How many complete cases are there?

Test data Count nulls/missing values in the test data:

```
sort(colSums(is.na(test_data)), decreasing = TRUE)
```

```
##           PH           MFR      Filler.Speed      Brand.Code
##          267            31           10            8
##      Fill.Ounces          PSC        PSC.CO2        PC.Volume
##           6            5            5            4
##      Carb.Pressure1  Hyd.Pressure4      PSC.Fill  Oxygen.Filler
##           4            4            3            3
##          Alch.Rel      Fill.Pressure      Filler.Level      Temperature
##           3            2            2            2
##      Usage.cont  Pressure.Setpoint      Carb.Rel      Carb.Volume
##           2            2            2            1
##      Carb.Temp      Hyd.Pressure2  Hyd.Pressure3      Density
##           1            1            1            1
##      Balling      Pressure.Vacuum  Bowl.Setpoint  Air.Pressurer
##           1            1            1            1
##      Carb.Pressure      Mnf.Flow      Hyd.Pressure1      Carb.Flow
##           0            0            0            0
##      Balling.Lvl
##           0
```

First off, all the **pH values are missing**. This is not a test set —it’s what we will predict. The training data will have to be split on its own into a training and test set.

Most columns are missing a few (under 10) values or none at all. MFR is missing the most values – 31 out of 267, or about 12%. Filler speed is next, missing only 10 values.

Next, we’ll look at the data row-wise and see if there are full or close-to-full missing cases.

```
rowSums(is.na(test_data))
```

```
##      [1]  3  1  1  4  1  2  1  4  1  2  3  1  1  1  1  4  1  1  1  1  1  2  3  1  1
##     [26]  1  1  1  2  1  1  2  2  2  2  1  1  1  1  2  1  2  1  1  1  1  1  1  1  1
##     [51]  3  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
##     [76]  1  2  1  1  1  1  1  1  1  1  5  2  1  1  1  1  2  15  1  2  2  1  3  1  2  1
##    [101]  2  1  1  1  1  1  1  1  1  1  2  1  1  1  1  3  1  1  2  1  1  2  1  1  1  1
##    [126]  1  2  2  1  1  2  2  2  3  1  1  3  2  2  2  1  1  1  2  1  1  1  1  1  2  1
##    [151]  1  1  1  1  1  1  1  1  1  1  1  2  2  1  1  1  2  2  1  1  1  2  2  1  1  1
##    [176]  1  1  1  2  1  1  1  1  1  1  2  1  1  1  1  1  1  1  1  2  1  1  1  1  1  1
##    [201]  1  1  2  1  1  1  1  2  2  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
##    [226]  1  3  1  1  1  1  2  1  1  1  2  1  1  1  1  1  1  2  1  1  1  1  1  3  1
##    [251]  2  1  3  1  1  9  2  1  1  1  1  1  1  1  1  1  1  1
```

```
#totals by row (1 does not count)
table(rowSums(is.na(test_data)))
```

```
##
##      1      2      3      4      5      9     15
##    200     50     11      3      1      1      1
```

```
#total NAs (subtracting 267 for the missing pH values)
sum(is.na(test_data)) - 267
```

```
## [1] 107
```

Overall, a data set with 267 rows and 33 columns is missing 107 values, or about 1.3%. 67 out of 267 cases are missing at least one value (about 25%).

One case has 14 missing values, so it may make sense to drop that; another is missing 8. Otherwise, most rows with missing values are missing only 1-3.

Training data Which variables are missing data?

```
sort(colSums(is.na(training_data)), decreasing = TRUE)
```

```
##           MFR           Brand.Code      Filler.Speed      PC.Volume
##           212             120           57             39
##      PSC.CO2      Fill.Ounces           PSC      Carb.Pressure1
##           39             38             33             32
##      Hyd.Pressure4      Carb.Pressure      Carb.Temp      PSC.Fill
##           30             27             26             23
##      Fill.Pressure      Filler.Level      Hyd.Pressure2      Hyd.Pressure3
##           22             20             15             15
##      Temperature      Oxygen.Filler      Pressure.Setpoint      Hyd.Pressure1
##           14             12             12             11
##      Carb.Volume      Carb.Rel           Alch.Rel           Usage.cont
##           10             10             9             5
##           PH           Mnf.Flow      Carb.Flow      Bowl.Setpoint
##           4             2             2             2
##      Density           Balling      Balling.Lvl      Pressure.Vacuum
##           1             1             1             0
##      Air.Pressurer
##           0
```

MFR is missing 212 out of 2571 cases, about 8.2%. Brand code, the only categorical variable, is missing 120 times, or about 4.7% of the time. Filler speed is missing 57 times, about 2.2%, and everything else is under 2%.

Let's see if anything strange is happening across rows:

```
#nothing out of the ordinary across rows. Most rows are complete.
#Rows with NAs typically 1 or 2 NAs, with
rowSums(is.na(training_data))
```

```
##      [1]  2  2  2  1  0  0  2  1  0  0  0  0  1  0  0  0  0  0  0  0  2  0  0
##     [25]  0  1  0  0  0  0  0  0  0  0  0  0  1  1  0  0  0  1  0  0  0  0  0  1
##     [49]  0  2  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0
##     [73]  0  0  0  0  0  0  0  2  0  0  0  0  0  0  0  0  0  0  0  0  2  0  0
##     [97]  0  1  0  0  0  0  0  0  0  0  0  0  2  5  0  0  0  0  0  1  0  0  0  0
##    [121]  0  0  0  0  2  0  0  1  0  0  0  0  0  1  0  0  0  2  0  0  0  0  0  0
##    [145]  0  0  0  0  0  0  1  1  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
##    [169]  0  0  0  0  0  0  0  0  1  0  2  0  0  0  0  0  2  0  0  0  0  0  0  0
##    [193]  0  0  0  1  0  0  0  0  0  0  0  0  0  1  1  1  1  1  1  1  0  0  0
##    [217]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  2  1  0  0  0  0  1
##    [241]  0  1  6  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  0  0  0  1  2
##    [265]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
```

```

## [289] 0 0 0 0 0 0 0 0 0 1 1 1 1 1 0 0 0 0 3 0 0 0 0 0 2
## [313] 0 0 0 0 0 0 0 0 1 0 0 0 1 1 0 1 0 0 0 1 0 0 1 2 0
## [337] 0 1 0 2 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0
## [361] 0 0 0 0 2 2 0 2 0 0 0 1 0 2 1 0 1 0 0 0 0 0 0 0 0
## [385] 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 2 0 0 0 0 0 0 0
## [409] 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 2 2 1 1 0
## [433] 0 0 2 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0
## [457] 0 0 1 0 0 0 1 0 0 0 0 0 0 0 2 1 0 0 4 1 1 1 1 1 1
## [481] 1 1 1 0 0 0 0 0 0 0 0 0 1 0 1 0 3 0 1 1 0 0 0 0 0
## [505] 0 0 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 0 0
## [529] 2 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 0 1 0 0 0 0 0 1 0
## [553] 0 0 0 0 0 1 0 1 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [577] 1 0 0 0 0 0 0 1 0 0 0 3 0 0 1 0 0 0 0 0 0 0 0 0 0
## [601] 0 0 0 0 0 0 0 0 2 4 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0
## [625] 0 0 0 0 3 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 0 0 0
## [649] 1 0 0 0 0 0 1 0 2 2 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0
## [673] 0 1 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0 5 2 0 0 1 2 3
## [697] 3 0 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 3 0
## [721] 0 2 0 0 0 0 0 0 0 0 0 0 1 2 0 0 0 0 9 4 0 0 0 0 0
## [745] 0 0 0 0 0 1 0 0 1 0 3 0 0 1 0 0 0 0 0 0 0 1 0 0 0
## [769] 0 0 2 2 3 3 3 3 4 3 3 3 3 3 3 3 4 0 0 0 0 0 0 0 0
## [793] 0 0 2 0 0 0 1 2 4 1 0 0 0 0 0 0 2 1 0 0 0 0 0 3
## [817] 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 2 1 0 0 0 1 0 0 0
## [841] 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [865] 2 0 0 0 0 0 0 0 0 0 0 0 0 0 2 2 0 0 0 0 1 0 0 0
## [889] 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 6 0 0 0
## [913] 2 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 0 0 0 0 0 1 0
## [937] 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 1 1 1 1 1 1
## [961] 2 1 1 2 2 0 0 0 1 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0
## [985] 0 0 0 0 0 0 0 0 0 0 2 1 1 1 0 3 0 0 0 0 0 0 0 0
## [1009] 0 0 0 0 0 2 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 2 0 0
## [1033] 0 0 2 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0
## [1057] 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0
## [1081] 0 0 0 0 0 0 0 1 0 0 0 0 0 8 2 0 1 0 0 0 0 1 0 0
## [1105] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1
## [1129] 0 0 0 0 0 0 0 0 3 1 0 0 0 2 0 0 0 0 0 0 0 0 0 0
## [1153] 0 0 0 0 1 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0
## [1177] 0 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 1 0 0 0 0 0 0
## [1201] 1 0 1 0 0 0 0 0 1 1 1 1 3 2 3 2 4 1 1 1 1 1 0 0
## [1225] 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0 0 0 4 2 0 0 1 0 0
## [1249] 0 0 0 0 3 0 0 0 0 0 0 2 0 0 0 0 1 0 0 1 1 1 0 0
## [1273] 1 0 0 0 0 0 1 3 0 0 0 0 1 0 0 0 0 0 0 0 2 0 2 1
## [1297] 0 1 0 4 0 1 0 1 0 0 1 1 1 0 0 0 0 0 1 1 1 2 1 1
## [1321] 1 1 1 2 1 1 0 0 0 0 1 1 1 1 1 1 2 1 1 1 2 1 1 1
## [1345] 1 1 2 1 1 1 1 1 1 1 2 1 2 0 1 1 1 1 1 1 3 1 1 2
## [1369] 2 1 0 0 0 0 0 2 0 0 0 0 0 0 0 1 1 1 1 0 0 0 0 0
## [1393] 0 0 0 2 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [1417] 0 0 0 0 3 2 0 0 0 0 0 1 0 0 3 0 0 0 0 0 0 0 0 0
## [1441] 0 0 0 0 0 0 0 0 3 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0
## [1465] 0 0 1 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0
## [1489] 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 1 0 0 0
## [1513] 0 0 0 0 0 0 0 2 0 0 0 0 0 0 0 4 1 0 0 0 0 0 6 0
## [1537] 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 1 1 1 1 1
## [1561] 2 3 0 0 0 0 0 0 0 0 0 0 0 1 3 0 0 0 0 0 0 0 0 0

```



```

## [1585] 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [1609] 0 1 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0
## [1633] 0 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [1657] 0 0 0 0 0 0 0 0 1 1 0 0 0 0 1 0 1 0 0 0 0 0 1
## [1681] 0 0 1 0 0 1 1 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1
## [1705] 0 0 1 0 0 0 0 0 0 0 0 0 0 0 12 1 0 0 1 0 3 1 2
## [1729] 0 0 0 1 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0
## [1753] 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0 0 0 1 1 0 1
## [1777] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 2 1
## [1801] 4 1 1 1 1 1 3 1 0 0 0 0 0 0 0 2 2 0 0 0 0 0 0
## [1825] 2 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 5 0 0
## [1849] 0 0 1 0 0 2 0 0 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0
## [1873] 0 0 0 0 0 0 3 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0
## [1897] 0 0 0 2 0 2 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 1
## [1921] 1 0 0 0 0 2 0 1 0 0 1 0 0 0 0 0 0 0 0 1 0 1 0
## [1945] 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 2 1 1 1 1 1 0
## [1969] 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0
## [1993] 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1
## [2017] 3 1 1 1 3 1 1 3 1 1 0 0 0 0 0 1 0 0 0 0 0 0 0
## [2041] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [2065] 0 2 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 2 0
## [2089] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 1 0
## [2113] 0 0 0 0 0 0 0 0 0 0 0 4 0 0 0 0 1 0 0 0 0 0 0
## [2137] 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 2 0 0
## [2161] 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0
## [2185] 1 1 1 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0
## [2209] 0 0 0 0 0 5 0 0 0 1 0 0 0 0 0 0 2 0 0 0 0 0 0
## [2233] 0 0 1 0 0 0 0 1 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0
## [2257] 2 1 1 1 0 0 0 0 0 0 0 0 0 0 4 0 0 0 0 0 0 2 0
## [2281] 0 1 0 0 0 0 1 0 0 0 0 2 0 0 0 0 0 1 0 0 0 2 0
## [2305] 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0 1 2
## [2329] 0 1 1 0 1 0 0 0 0 0 1 2 0 0 0 0 0 0 0 0 0 0 0
## [2353] 0 0 0 0 2 0 3 0 0 0 0 0 0 0 0 0 1 0 0 3 0 0 0
## [2377] 0 0 0 2 0 0 0 0 2 3 0 0 1 0 0 0 0 0 0 1 0 0 0
## [2401] 0 0 0 0 0 0 0 0 1 0 0 0 2 2 0 0 0 0 0 0 0 0 0
## [2425] 0 0 1 1 1 0 0 0 0 0 0 2 0 1 0 0 0 0 0 0 0 1 0
## [2449] 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0 1 1
## [2473] 0 0 0 0 2 0 2 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0
## [2497] 0 0 0 0 0 0 0 1 0 0 0 7 1 0 0 0 0 2 0 0 0 0 0
## [2521] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0
## [2545] 0 0 0 0 0 1 0 0 0 0 0 0 0 0 2 0 0 0 0 1 0 0 1
## [2569] 0 0 3

```

```
table(rowSums(is.na(training_data)))
```

```

##
##      0      1      2      3      4      5      6      7      8      9     12
## 2038 345 119  45  13   4   3   1   1   1   1

```

```
sum(is.na(test_data))
```

```
## [1] 374
```

Overall, 374 values are missing (about 14.5% total missingness). 2038/2571 cases are complete. That's about 79%; 21% of cases are incomplete. Let's also check to see if the data is MCAR.

```
library(naniar)
numeric_only <- training_data |>
  select(-Brand.Code)

numeric_only <- numeric_only |>
  dplyr::select(where(is.numeric)) |>
  mutate(across(everything(), as.double))

mcar_test(numeric_only)
```

```
## # A tibble: 1 x 4
##   statistic    df p.value missing.patterns
##   <dbl> <dbl> <dbl>         <int>
## 1   10791.  2871      0             99
```

The p-value is 0, which means the data most likely isn't missing completely at random (there are patterns in the missing data).

Data cleaning

We'll cover:

- Variables with zero or near-zero variance
- Highly correlated variables
- Outliers/range of variables/scaling (if needed)

Checking for variables with near-zero or zero variance

This data set has 33 variables. Let's check to see if they're all (potentially) providing useful predictive information by testing for near-zero or zero variance.

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.5.2
```

```
## Loading required package: lattice
```

```
##
```

```
## Attaching package: 'caret'
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
## lift
```

```
nzv_results <- nearZeroVar(training_data, saveMetrics = TRUE)
```

```
nzv_results
```

##	freqRatio	percentUnique	zeroVar	nzv
## Brand.Code	2.014634	0.1555815	FALSE	FALSE
## Carb.Volume	1.055556	3.9284325	FALSE	FALSE
## Fill.Ounces	1.163043	3.5783742	FALSE	FALSE
## PC.Volume	1.100000	17.6584986	FALSE	FALSE
## Carb.Pressure	1.016393	4.1229094	FALSE	FALSE
## Carb.Temp	1.016667	4.7841307	FALSE	FALSE
## PSC	1.211538	5.0175029	FALSE	FALSE
## PSC.Fill	1.091892	1.2446519	FALSE	FALSE
## PSC.CO2	1.078303	0.5056398	FALSE	FALSE
## Mnf.Flow	1.048443	18.9420459	FALSE	FALSE
## Carb.Pressure1	1.031746	5.4453520	FALSE	FALSE
## Fill.Pressure	1.762931	4.2007001	FALSE	FALSE
## Hyd.Pressure1	31.111111	9.5293660	FALSE	TRUE
## Hyd.Pressure2	7.271028	8.0513419	FALSE	FALSE
## Hyd.Pressure3	11.450704	7.4679113	FALSE	FALSE
## Hyd.Pressure4	1.008264	1.5558149	FALSE	FALSE
## Filler.Level	1.086957	11.2018670	FALSE	FALSE
## Filler.Speed	1.045161	9.4904706	FALSE	FALSE
## Temperature	1.040000	2.1781408	FALSE	FALSE
## Usage.cont	1.105263	18.7086737	FALSE	FALSE
## Carb.Flow	1.586207	20.7312330	FALSE	FALSE
## Density	1.108374	3.0338390	FALSE	FALSE
## MFR	1.222222	22.8315830	FALSE	FALSE
## Balling	1.193182	8.4402956	FALSE	FALSE
## Pressure.Vacuum	1.389728	0.6223259	FALSE	FALSE
## PH	1.078125	2.0225593	FALSE	FALSE
## Oxygen.Filler	1.266667	13.1466356	FALSE	FALSE
## Bowl.Setpoint	2.990847	0.4278491	FALSE	FALSE
## Pressure.Setpoint	1.319361	0.3111630	FALSE	FALSE
## Air.Pressurer	1.093407	1.2446519	FALSE	FALSE
## Alch.Rel	1.098315	2.0614547	FALSE	FALSE
## Carb.Rel	1.011583	1.6336056	FALSE	FALSE
## Balling.Lvl	1.294872	3.1894205	FALSE	FALSE

Hyd.Pressure1 has a near-zero variance, so we could eliminate that one.

Looking for highly correlated variables

We'll create a correlation grid, so we can see actual correlation between variables. We'll also create a correlation plot and use findCorrelation from the caret package to find variables that have 90% or higher correlation.

```
correlation_grid <- training_data |>
  drop_na() |>
  select(-Brand.Code) |>
  cor()
```

```
correlation_grid <- as.data.frame(correlation_grid)
```

```
correlation_grid
```

##	Carb.Volume	Fill.Ounces	PC.Volume	Carb.Pressure
## Carb.Volume	1.000000000	0.022648036	-0.243234517	0.4278586099
## Fill.Ounces	0.022648036	1.000000000	-0.172701966	0.0044669914
## PC.Volume	-0.243234517	-0.172701966	1.000000000	-0.1300941804
## Carb.Pressure	0.427858610	0.004466991	-0.130094180	1.0000000000
## Carb.Temp	-0.135080260	-0.020073097	0.008599427	0.8243350333
## PSC	-0.020625187	0.043056041	0.190961266	-0.0520086907
## PSC.Fill	-0.012364141	0.069748653	-0.055914847	-0.0179275381
## PSC.CO2	-0.051977547	0.014705610	-0.055356933	-0.0174011136
## Mnf.Flow	0.100525364	-0.005787251	-0.203032882	0.0189865815
## Carb.Pressure1	0.081146212	-0.004186326	-0.238496181	0.0221844474
## Fill.Pressure	-0.096123314	0.068377098	-0.095744421	-0.0850458606
## Hyd.Pressure1	-0.051894268	-0.146359721	0.271171741	-0.0704043564
## Hyd.Pressure2	0.033567314	-0.143047986	0.028768135	-0.0267152675
## Hyd.Pressure3	0.057610071	-0.107816452	-0.024741466	-0.0082140513
## Hyd.Pressure4	-0.567908627	0.056086557	0.132413786	-0.3229128556
## Filler.Level	-0.018377226	0.010856355	0.220434988	0.0097134297
## Filler.Speed	0.003233452	0.044277046	-0.073766021	0.0320230919
## Temperature	-0.189760851	-0.011208233	0.142065435	-0.0490025241
## Usage.cont	0.073297402	0.085289046	-0.269204705	0.0004415267
## Carb.Flow	-0.088782919	-0.116991410	0.254535534	-0.0111938869
## Density	0.801808803	-0.080414770	-0.167310345	0.4591743002
## MFR	0.010144634	0.043445999	-0.066678296	0.0292986178
## Balling	0.818778791	-0.068439337	-0.204785622	0.4539307252
## Pressure.Vacuum	-0.079190180	0.052165821	-0.051605143	0.0051279147
## PH	0.073251595	-0.118207504	0.097746859	0.0788617427
## Oxygen.Filler	-0.112682435	-0.032064258	0.212285069	-0.0506609705
## Bowl.Setpoint	0.001993607	0.011188522	0.210972310	0.0187581570
## Pressure.Setpoint	-0.160640146	0.065078231	-0.022210124	-0.1171228371
## Air.Pressurer	-0.024854172	0.058101103	-0.036605483	0.0219237613
## Alch.Rel	0.824787575	-0.110432476	-0.177865067	0.4590380468
## Carb.Rel	0.841072250	-0.118666586	-0.167433257	0.4736224720
## Balling.Lvl	0.817993347	-0.063820071	-0.210081156	0.4589466196
##	Carb.Temp	PSC	PSC.Fill	PSC.CO2
## Carb.Volume	-0.135080260	-0.020625187	-0.012364141	-0.051977547
## Fill.Ounces	-0.020073097	0.043056041	0.069748653	0.014705610
## PC.Volume	0.008599427	0.190961266	-0.055914847	-0.055356933
## Carb.Pressure	0.824335033	-0.052008691	-0.017927538	-0.017401114
## Carb.Temp	1.000000000	-0.044380293	-0.010968792	0.023660514
## PSC	-0.044380293	1.000000000	0.193242059	0.053922116
## PSC.Fill	-0.010968792	0.193242059	1.000000000	0.201967429
## PSC.CO2	0.023660514	0.053922116	0.201967429	1.000000000
## Mnf.Flow	-0.030110321	0.042982180	-0.033801564	0.049931986
## Carb.Pressure1	-0.014312694	-0.011071066	-0.029772433	0.038316144
## Fill.Pressure	-0.028433189	0.030908100	-0.010528126	0.078519840
## Hyd.Pressure1	-0.045431217	-0.019758828	-0.053220514	-0.049037869
## Hyd.Pressure2	-0.039543090	-0.033813839	-0.080467321	-0.021838338
## Hyd.Pressure3	-0.035120030	-0.020448188	-0.068958062	0.005757630
## Hyd.Pressure4	-0.002573081	0.036485173	0.019810466	0.054940092

## Filler.Level	0.011038984	-0.016061814	0.048227359	-0.042306839
## Filler.Speed	0.029355933	-0.002395360	-0.017295907	-0.008421406
## Temperature	0.061914927	0.014217909	0.030080929	0.037007308
## Usage.cont	-0.043584359	0.049860720	-0.025727952	0.032502663
## Carb.Flow	0.045050619	-0.037361416	0.017579095	-0.013058433
## Density	0.021593836	-0.067168725	-0.024708871	-0.045382139
## MFR	0.023442657	0.007659111	-0.008663604	-0.004263878
## Balling	0.006005105	-0.064450826	-0.019108545	-0.043036807
## Pressure.Vacuum	0.037327836	0.055246194	0.033093793	-0.018438819
## PH	0.034594375	-0.069906559	-0.031374041	-0.088759922
## Oxygen.Filler	0.012010201	0.002765929	-0.005769535	-0.016971816
## Bowl.Setpoint	0.007550616	-0.020808347	0.049191096	-0.044820225
## Pressure.Setpoint	-0.027029545	0.042979144	-0.005800041	0.082772777
## Air.Pressurer	0.038177076	0.061433424	-0.017995701	-0.006984004
## Alch.Rel	0.004966509	-0.060809035	-0.017045886	-0.050859393
## Carb.Rel	0.009020802	-0.065354969	-0.015365172	-0.066475871
## Balling.Lvl	0.010458674	-0.061576860	-0.016064731	-0.047296997
##	Mnf.Flow	Carb.Pressure1	Fill.Pressure	Hyd.Pressure1
## Carb.Volume	0.100525364	0.081146212	-0.09612331	-0.051894268
## Fill.Ounces	-0.005787251	-0.004186326	0.06837710	-0.146359721
## PC.Volume	-0.203032882	-0.238496181	-0.09574442	0.271171741
## Carb.Pressure	0.018986582	0.022184447	-0.08504586	-0.070404356
## Carb.Temp	-0.030110321	-0.014312694	-0.02843319	-0.045431217
## PSC	0.042982180	-0.011071066	0.03090810	-0.019758828
## PSC.Fill	-0.033801564	-0.029772433	-0.01052813	-0.053220514
## PSC.CO2	0.049931986	0.038316144	0.07851984	-0.049037869
## Mnf.Flow	1.000000000	0.465355315	0.49111329	0.298997690
## Carb.Pressure1	0.465355315	1.000000000	0.22002705	-0.012711716
## Fill.Pressure	0.491113286	0.220027054	1.00000000	0.137594249
## Hyd.Pressure1	0.298997690	-0.012711716	0.13759425	1.000000000
## Hyd.Pressure2	0.648337926	0.284710255	0.31836940	0.711851345
## Hyd.Pressure3	0.764142833	0.375658047	0.43637028	0.605589019
## Hyd.Pressure4	0.073124200	0.038936494	0.29144885	0.051658157
## Filler.Level	-0.608380668	-0.425646963	-0.45892889	-0.029684260
## Filler.Speed	0.024363769	0.012389362	-0.21347499	-0.022850088
## Temperature	-0.063477350	-0.070067643	0.06397894	-0.073341745
## Usage.cont	0.560192617	0.354875344	0.30257079	0.118691986
## Carb.Flow	-0.497211539	-0.273632871	-0.18715299	-0.169136801
## Density	0.021343082	0.040888366	-0.20887500	-0.003601856
## MFR	0.019593355	0.013931715	-0.21290076	-0.035646540
## Balling	0.112393982	0.073508154	-0.15676227	0.025863408
## Pressure.Vacuum	-0.559943315	-0.284881843	-0.30810832	-0.324133005
## PH	-0.457218313	-0.112108310	-0.31311751	-0.048631533
## Oxygen.Filler	-0.550066399	-0.278795209	-0.24769592	-0.118317983
## Bowl.Setpoint	-0.611580420	-0.432671459	-0.42609290	-0.029844050
## Pressure.Setpoint	0.482530739	0.234201877	0.81528556	0.190079720
## Air.Pressurer	-0.041525175	0.061506659	0.03280945	-0.209227461
## Alch.Rel	0.027999719	0.009047055	-0.20467008	0.006667887
## Carb.Rel	-0.027880871	0.004804776	-0.20272201	0.034387945
## Balling.Lvl	0.042267589	0.033601804	-0.17885789	-0.007754220
##	Hyd.Pressure2	Hyd.Pressure3	Hyd.Pressure4	Filler.Level
## Carb.Volume	0.03356731	0.057610071	-0.567908627	-0.01837723
## Fill.Ounces	-0.14304799	-0.107816452	0.056086557	0.01085635
## PC.Volume	0.02876814	-0.024741466	0.132413786	0.22043499

## Carb.Pressure	-0.02671527	-0.008214051	-0.322912856	0.00971343
## Carb.Temp	-0.03954309	-0.035120030	-0.002573081	0.01103898
## PSC	-0.03381384	-0.020448188	0.036485173	-0.01606181
## PSC.Fill	-0.08046732	-0.068958062	0.019810466	0.04822736
## PSC.CO2	-0.02183834	0.005757630	0.054940092	-0.04230684
## Mnf.Flow	0.64833793	0.764142833	0.073124200	-0.60838067
## Carb.Pressure1	0.28471026	0.375658047	0.038936494	-0.42564696
## Fill.Pressure	0.31836940	0.436370279	0.291448848	-0.45892889
## Hyd.Pressure1	0.71185135	0.605589019	0.051658157	-0.02968426
## Hyd.Pressure2	1.00000000	0.917661875	0.022980025	-0.41103521
## Hyd.Pressure3	0.91766187	1.000000000	0.046649319	-0.52796590
## Hyd.Pressure4	0.02298003	0.046649319	1.000000000	-0.10010344
## Filler.Level	-0.41103521	-0.527965898	-0.100103442	1.00000000
## Filler.Speed	0.07028157	0.031692930	-0.259685780	0.06319357
## Temperature	-0.13725053	-0.120288087	0.283933984	0.05040323
## Usage.cont	0.37224053	0.419761856	0.070916924	-0.35740823
## Carb.Flow	-0.32187768	-0.363691033	-0.009527157	0.01743320
## Density	0.05842429	0.040354370	-0.584381995	0.00772950
## MFR	0.04937645	0.012841039	-0.280014344	0.06935680
## Balling	0.10480326	0.125869294	-0.594996647	0.01188558
## Pressure.Vacuum	-0.62816754	-0.672258326	-0.054652623	0.35866815
## PH	-0.21571713	-0.260535183	-0.184518087	0.34879540
## Oxygen.Filler	-0.29438355	-0.363660021	0.009692254	0.22540536
## Bowl.Setpoint	-0.41457787	-0.529215855	-0.117019292	0.97724166
## Pressure.Setpoint	0.34183180	0.459918732	0.288770453	-0.44595292
## Air.Pressurer	-0.17237591	-0.067700529	0.021731381	-0.12573419
## Alch.Rel	0.03916606	0.049948952	-0.685576013	0.04355856
## Carb.Rel	0.03150549	0.013543248	-0.581827500	0.13107856
## Balling.Lvl	0.02885940	0.041595480	-0.575044299	0.05317923
##	Filler.Speed	Temperature	Usage.cont	Carb.Flow
## Carb.Volume	0.003233452	-0.18976085	0.0732974017	-0.088782919
## Fill.Ounces	0.044277046	-0.01120823	0.0852890456	-0.116991410
## PC.Volume	-0.073766021	0.14206544	-0.2692047046	0.254535534
## Carb.Pressure	0.032023092	-0.04900252	0.0004415267	-0.011193887
## Carb.Temp	0.029355933	0.06191493	-0.0435843590	0.045050619
## PSC	-0.002395360	0.01421791	0.0498607203	-0.037361416
## PSC.Fill	-0.017295907	0.03008093	-0.0257279520	0.017579095
## PSC.CO2	-0.008421406	0.03700731	0.0325026634	-0.013058433
## Mnf.Flow	0.024363769	-0.06347735	0.5601926171	-0.497211539
## Carb.Pressure1	0.012389362	-0.07006764	0.3548753438	-0.273632871
## Fill.Pressure	-0.213474995	0.06397894	0.3025707885	-0.187152994
## Hyd.Pressure1	-0.022850088	-0.07334175	0.1186919862	-0.169136801
## Hyd.Pressure2	0.070281568	-0.13725053	0.3722405332	-0.321877677
## Hyd.Pressure3	0.031692930	-0.12028809	0.4197618556	-0.363691033
## Hyd.Pressure4	-0.259685780	0.28393398	0.0709169237	-0.009527157
## Filler.Level	0.063193573	0.05040323	-0.3574082260	0.017433196
## Filler.Speed	1.000000000	-0.06794451	0.0466749382	-0.062641899
## Temperature	-0.067944506	1.000000000	-0.0950428265	0.140119431
## Usage.cont	0.046674938	-0.09504283	1.0000000000	-0.488854072
## Carb.Flow	-0.062641899	0.14011943	-0.4888540721	1.000000000
## Density	0.025454377	-0.19369890	-0.0045531798	0.023058872
## MFR	0.951264445	-0.08156825	0.0397443802	-0.074148067
## Balling	0.037938064	-0.23844781	0.0692694347	-0.102299824
## Pressure.Vacuum	0.045187804	0.04313148	-0.3196915871	0.290026894

## PH	-0.047596639	-0.18152910	-0.3577042377	0.240757744
## Oxygen.Filler	-0.044126285	0.09338308	-0.3164959408	0.385692082
## Bowl.Setpoint	0.027688288	0.05010937	-0.3584563273	0.012221618
## Pressure.Setpoint	-0.068525700	0.05904194	0.2626733751	-0.166362628
## Air.Pressurer	-0.006199674	0.06692883	-0.1025056258	0.137849013
## Alch.Rel	-0.005598672	-0.24776232	-0.0237408078	0.007527013
## Carb.Rel	0.005300760	-0.13595194	-0.0390798757	-0.006940815
## Balling.Lvl	0.003823258	-0.22497504	0.0232069975	-0.055365859
##	Density	MFR	Balling	Pressure.Vacuum
## Carb.Volume	0.801808803	0.010144634	0.818778791	-0.079190180
## Fill.Ounces	-0.080414770	0.043445999	-0.068439337	0.052165821
## PC.Volume	-0.167310345	-0.066678296	-0.204785622	-0.051605143
## Carb.Pressure	0.459174300	0.029298618	0.453930725	0.005127915
## Carb.Temp	0.021593836	0.023442657	0.006005105	0.037327836
## PSC	-0.067168725	0.007659111	-0.064450826	0.055246194
## PSC.Fill	-0.024708871	-0.008663604	-0.019108545	0.033093793
## PSC.CO2	-0.045382139	-0.004263878	-0.043036807	-0.018438819
## Mnf.Flow	0.021343082	0.019593355	0.112393982	-0.559943315
## Carb.Pressure1	0.040888366	0.013931715	0.073508154	-0.284881843
## Fill.Pressure	-0.208875002	-0.212900763	-0.156762274	-0.308108315
## Hyd.Pressure1	-0.003601856	-0.035646540	0.025863408	-0.324133005
## Hyd.Pressure2	0.058424287	0.049376448	0.104803262	-0.628167539
## Hyd.Pressure3	0.040354370	0.012841039	0.125869294	-0.672258326
## Hyd.Pressure4	-0.584381995	-0.280014344	-0.594996647	-0.054652623
## Filler.Level	0.007729500	0.069356797	0.011885578	0.358668150
## Filler.Speed	0.025454377	0.951264445	0.037938064	0.045187804
## Temperature	-0.193698895	-0.081568252	-0.238447810	0.043131483
## Usage.cont	-0.004553180	0.039744380	0.069269435	-0.319691587
## Carb.Flow	0.023058872	-0.074148067	-0.102299824	0.290026894
## Density	1.000000000	0.039383665	0.953127632	-0.083762802
## MFR	0.039383665	1.000000000	0.051171065	0.043987724
## Balling	0.953127632	0.051171065	1.000000000	-0.167052546
## Pressure.Vacuum	-0.083762802	0.043987724	-0.167052546	1.000000000
## PH	0.095340535	-0.052377217	0.072297357	0.213173584
## Oxygen.Filler	-0.046621720	-0.048876320	-0.113757056	0.278478283
## Bowl.Setpoint	0.019792536	0.042312091	0.027415802	0.358696671
## Pressure.Setpoint	-0.250884909	-0.064390559	-0.210098742	-0.298520301
## Air.Pressurer	-0.082976446	-0.008182360	-0.102307854	0.175090738
## Alch.Rel	0.922993173	-0.001003949	0.945725964	-0.054788103
## Carb.Rel	0.860322233	0.010884909	0.858990647	-0.008434181
## Balling.Lvl	0.956417297	0.016039036	0.988412082	-0.051626926
##	PH	Oxygen.Filler	Bowl.Setpoint	Pressure.Setpoint
## Carb.Volume	0.073251595	-0.112682435	0.001993607	-0.160640146
## Fill.Ounces	-0.118207504	-0.032064258	0.011188522	0.065078231
## PC.Volume	0.097746859	0.212285069	0.210972310	-0.022210124
## Carb.Pressure	0.078861743	-0.050660970	0.018758157	-0.117122837
## Carb.Temp	0.034594375	0.012010201	0.007550616	-0.027029545
## PSC	-0.069906559	0.002765929	-0.020808347	0.042979144
## PSC.Fill	-0.031374041	-0.005769535	0.049191096	-0.005800041
## PSC.CO2	-0.088759922	-0.016971816	-0.044820225	0.082772777
## Mnf.Flow	-0.457218313	-0.550066399	-0.611580420	0.482530739
## Carb.Pressure1	-0.112108310	-0.278795209	-0.432671459	0.234201877
## Fill.Pressure	-0.313117513	-0.247695921	-0.426092896	0.815285564
## Hyd.Pressure1	-0.048631533	-0.118317983	-0.029844050	0.190079720

## Hyd.Pressure2	-0.215717133	-0.294383550	-0.414577866	0.341831797
## Hyd.Pressure3	-0.260535183	-0.363660021	-0.529215855	0.459918732
## Hyd.Pressure4	-0.184518087	0.009692254	-0.117019292	0.288770453
## Filler.Level	0.348795399	0.225405359	0.977241656	-0.445952917
## Filler.Speed	-0.047596639	-0.044126285	0.027688288	-0.068525700
## Temperature	-0.181529097	0.093383081	0.050109366	0.059041937
## Usage.cont	-0.357704238	-0.316495941	-0.358456327	0.262673375
## Carb.Flow	0.240757744	0.385692082	0.012221618	-0.166362628
## Density	0.095340535	-0.046621720	0.019792536	-0.250884909
## MFR	-0.052377217	-0.048876320	0.042312091	-0.064390559
## Balling	0.072297357	-0.113757056	0.027415802	-0.210098742
## Pressure.Vacuum	0.213173584	0.278478283	0.358696671	-0.298520301
## PH	1.000000000	0.162016479	0.358504933	-0.316950863
## Oxygen.Filler	0.162016479	1.000000000	0.224734144	-0.241221158
## Bowl.Setpoint	0.358504933	0.224734144	1.000000000	-0.440555036
## Pressure.Setpoint	-0.316950863	-0.241221158	-0.440555036	1.000000000
## Air.Pressurer	-0.008276544	0.116495267	-0.130939728	0.073805550
## Alch.Rel	0.156053375	-0.050111016	0.063332637	-0.268800293
## Carb.Rel	0.189575611	-0.039206163	0.152117160	-0.261257572
## Balling.Lvl	0.103279163	-0.074544984	0.070442468	-0.242343282
##	Air.Pressurer	Alch.Rel	Carb.Rel	Balling.Lvl
## Carb.Volume	-0.024854172	0.824787575	0.841072250	0.817993347
## Fill.Ounces	0.058101103	-0.110432476	-0.118666586	-0.063820071
## PC.Volume	-0.036605483	-0.177865067	-0.167433257	-0.210081156
## Carb.Pressure	0.021923761	0.459038047	0.473622472	0.458946620
## Carb.Temp	0.038177076	0.004966509	0.009020802	0.010458674
## PSC	0.061433424	-0.060809035	-0.065354969	-0.061576860
## PSC.Fill	-0.017995701	-0.017045886	-0.015365172	-0.016064731
## PSC.CO2	-0.006984004	-0.050859393	-0.066475871	-0.047296997
## Mnf.Flow	-0.041525175	0.027999719	-0.027880871	0.042267589
## Carb.Pressure1	0.061506659	0.009047055	0.004804776	0.033601804
## Fill.Pressure	0.032809452	-0.204670077	-0.202722006	-0.178857893
## Hyd.Pressure1	-0.209227461	0.006667887	0.034387945	-0.007754220
## Hyd.Pressure2	-0.172375910	0.039166055	0.031505491	0.028859401
## Hyd.Pressure3	-0.067700529	0.049948952	0.013543248	0.041595480
## Hyd.Pressure4	0.021731381	-0.685576013	-0.581827500	-0.575044299
## Filler.Level	-0.125734192	0.043558564	0.131078561	0.053179225
## Filler.Speed	-0.006199674	-0.005598672	0.005300760	0.003823258
## Temperature	0.066928829	-0.247762315	-0.135951942	-0.224975043
## Usage.cont	-0.102505626	-0.023740808	-0.039079876	0.023206997
## Carb.Flow	0.137849013	0.007527013	-0.006940815	-0.055365859
## Density	-0.082976446	0.922993173	0.860322233	0.956417297
## MFR	-0.008182360	-0.001003949	0.010884909	0.016039036
## Balling	-0.102307854	0.945725964	0.858990647	0.988412082
## Pressure.Vacuum	0.175090738	-0.054788103	-0.008434181	-0.051626926
## PH	-0.008276544	0.156053375	0.189575611	0.103279163
## Oxygen.Filler	0.116495267	-0.050111016	-0.039206163	-0.074544984
## Bowl.Setpoint	-0.130939728	0.063332637	0.152117160	0.070442468
## Pressure.Setpoint	0.073805550	-0.268800293	-0.261257572	-0.242343282
## Air.Pressurer	1.000000000	-0.082451451	-0.107386102	-0.085434755
## Alch.Rel	-0.082451451	1.000000000	0.882791450	0.947343341
## Carb.Rel	-0.107386102	0.882791450	1.000000000	0.872148303
## Balling.Lvl	-0.085434755	0.947343341	0.872148303	1.000000000

There are many highly correlated variables. A quick scan shows:

- hyd.pressure 2 and 3 have a .91 correlation
- filler.level and bowl.set.point have a .98 correlation
- fillers.speed and MFR have a .95 correlation
- density and balling.lvl have a .95 correlation
- balling and balling.lvl have a .98 correlation; both are strongly correlated with density and alch rel.
- alch.rel and balling.lvl have a .95 correlation; it also has a .95 correlation with balling, balling.lvl, and .93 with density

Here's a correlation plot so we can see all the positively and negatively correlated variables:

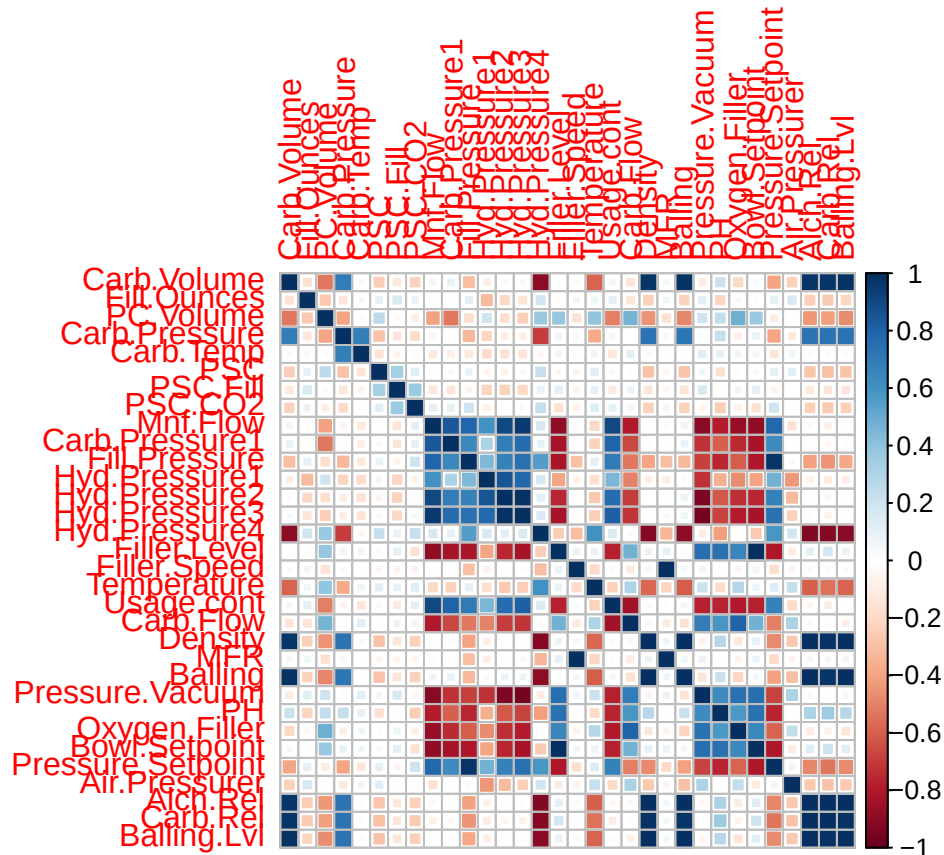
```
library(corrplot)
```

```
## Warning: package 'corrplot' was built under R version 4.5.3
```

```
## corrplot 0.95 loaded
```

```
grid <- training_data |>
  drop_na() |>
  select(-Brand.Code) |>
  cor()

corrplot(cor(grid),method = "square")
```



There are many negatively correlated variables here that do not need to be removed, but may be helpful for analysis. We can see that the response variable, pH, has a strong negative correlation with some variables, including fill.pressure1, usage.cont, and Air.Pressurer.

The findCorrelation feature will decide which variables to drop (it leaves the variable that best represents the group):

```
library(caret)

#if we remove colinear variables at 95% correlation, we would drop 10 variables
indexestodrop95 <- findCorrelation(cor(grid), cutoff = 0.95)
colnames(grid)[indexestodrop95]
```

```
## [1] "Pressure.Setpoint" "Mnf.Flow"          "Hyd.Pressure3"
## [4] "Bowl.Setpoint"     "Carb.Rel"          "Balling"
## [7] "Alch.Rel"          "Balling.Lvl"       "Density"
## [10] "Filler.Speed"
```

#this will leave us with 22 predictor variables and the response variable.

```
#if we remove colinear variables at 90% correlation, we would drop 12 variables
indexestodrop90 <- findCorrelation(cor(grid), cutoff = 0.90)
colnames(grid)[indexestodrop90]
```

```
## [1] "Pressure.Setpoint" "Mnf.Flow"          "Hyd.Pressure3"
## [4] "Pressure.Vacuum"   "Hyd.Pressure4"     "Bowl.Setpoint"
## [7] "Carb.Rel"          "Balling"           "Alch.Rel"
## [10] "Balling.Lvl"       "Density"           "Filler.Speed"
```

#this will leave us with 20 predictor variables and the response variable.

Removing variables

Removing some of the 33 variables will help simplify this data set. We will remove the variable with NZV and highly colinear variables (greater than or about .90, as defined above).

the findcorrelation function says to drop Filler.Speed instead of MFR. However, we will opt to drop MFR instead, given that it has the highest number of missing values (filler.speed has 57 as opposed to MFR's 212). Having to impute all these values will makes the variable less reliable, and dropping it would mean having to drop more cases.

```
#removing the variable with NZV
training_data_new <- training_data |>
  select(-Hyd.Pressure1)

#removing correlated variables
training_data_new <- training_data_new |>
  select(-Pressure.Setpoint, -MFR, -Hyd.Pressure3,
        -Pressure.Vacuum, -Hyd.Pressure4, -Bowl.Setpoint, -Carb.Rel,
        -Balling, -Alch.Rel, -Balling.Lvl, -Density)

#donig the same for the test set
```

```
test_data_new <- test_data |>
  select(-Hyd.Pressure1, -Pressure.Setpoint, -MFR, -Hyd.Pressure3,
         -Pressure.Vacuum, -Hyd.Pressure4, -Bowl.Setpoint, -Carb.Rel,
         -Balling, -Alch.Rel, -Balling.Lvl, -Density)
```

Checking for outliers

Before we deal with missing values, let's see if there are any glaring outliers or data entry errors.

#may be an outlier in oxygen filler - the max values is much higher than the

```
summary(test_data_new)
```

```
## Brand.Code      Carb.Volume      Fill.Ounces      PC.Volume
## Length:267      Min.      :5.147      Min.      :23.75      Min.      :0.09867
## Class :character 1st Qu.:5.287      1st Qu.:23.92      1st Qu.:0.23333
## Mode  :character Median :5.340      Median :23.97      Median :0.27533
##                Mean  :5.369      Mean  :23.97      Mean  :0.27769
##                3rd Qu.:5.465      3rd Qu.:24.01      3rd Qu.:0.32200
##                Max.  :5.667      Max.  :24.20      Max.  :0.46400
##                NA's   :1         NA's   :6         NA's   :4
## Carb.Pressure    Carb.Temp      PSC              PSC.Fill
## Min.      :60.20      Min.      :130.0      Min.      :0.00400      Min.      :0.0200
## 1st Qu.:65.30      1st Qu.:138.4      1st Qu.:0.04450      1st Qu.:0.1000
## Median :68.00      Median :140.8      Median :0.07600      Median :0.1800
## Mean  :68.25      Mean  :141.2      Mean  :0.08545      Mean  :0.1903
## 3rd Qu.:70.60      3rd Qu.:143.8      3rd Qu.:0.11200      3rd Qu.:0.2600
## Max.  :77.60      Max.  :154.0      Max.  :0.24600      Max.  :0.6200
##                NA's   :1         NA's   :5         NA's   :3
## PSC.CO2          Mnf.Flow      Carb.Pressure1    Fill.Pressure
## Min.      :0.00000      Min.      :-100.20      Min.      :113.0      Min.      :37.80
## 1st Qu.:0.02000      1st Qu.: -100.00      1st Qu.:120.2      1st Qu.:46.00
## Median :0.04000      Median :    0.20      Median :123.4      Median :47.80
## Mean  :0.05107      Mean  :   21.03      Mean  :123.0      Mean  :48.14
## 3rd Qu.:0.06000      3rd Qu.: 141.30      3rd Qu.:125.5      3rd Qu.:50.20
## Max.  :0.24000      Max.  : 220.40      Max.  :136.0      Max.  :60.20
## NA's   :5              NA's   :4              NA's   :2
## Hyd.Pressure2    Filler.Level    Filler.Speed      Temperature
## Min.      :-50.00      Min.      : 69.2      Min.      :1006      Min.      :63.80
## 1st Qu.:  0.00      1st Qu.:100.6      1st Qu.:3812      1st Qu.:65.40
## Median : 26.80      Median :118.6      Median :3978      Median :65.80
## Mean  : 20.11      Mean  :110.3      Mean  :3581      Mean  :66.23
## 3rd Qu.: 34.80      3rd Qu.:120.2      3rd Qu.:3996      3rd Qu.:66.60
## Max.  : 61.40      Max.  :153.2      Max.  :4020      Max.  :75.40
## NA's   :1          NA's   :2          NA's   :10         NA's   :2
## Usage.cont      Carb.Flow      PH              Oxygen.Filler
## Min.      :12.90      Min.      :  0      Mode:logical      Min.      :0.00240
## 1st Qu.:18.12      1st Qu.:1083      NA's:267          1st Qu.:0.01960
## Median :21.44      Median :3038              Median :0.03370
## Mean  :20.90      Mean  :2409              Mean  :0.04666
## 3rd Qu.:23.74      3rd Qu.:3215              3rd Qu.:0.05440
## Max.  :24.60      Max.  :3858              Max.  :0.39800
```

```
## NA's :2
## Air.Pressurer
## Min. :141.2
## 1st Qu.:142.2
## Median :142.6
## Mean :142.8
## 3rd Qu.:142.8
## Max. :147.2
## NA's :1
```

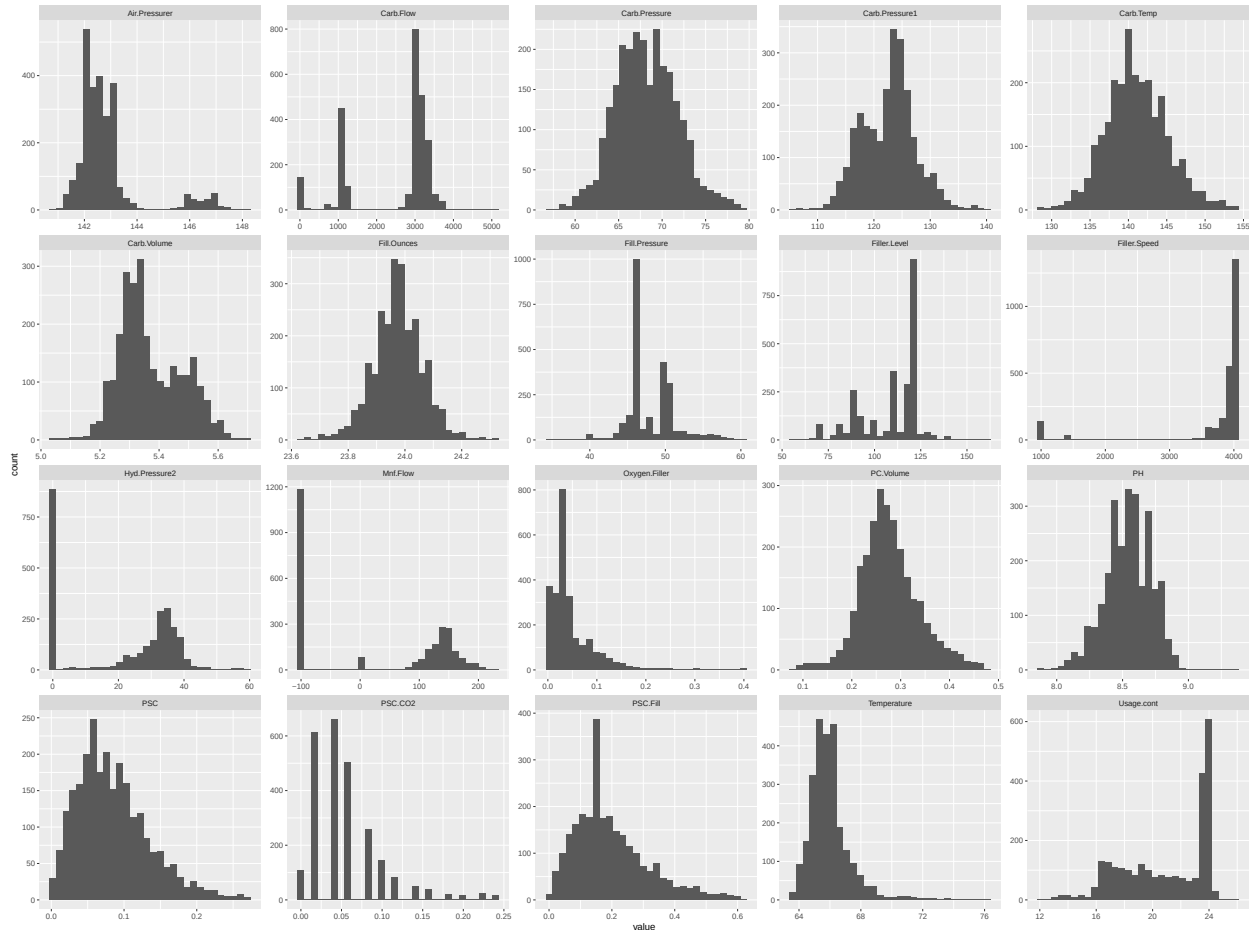
#there's also at least one really large value on oxygen filler here
summary(training_data_new)

```
## Brand.Code Carb.Volume Fill.Ounces PC.Volume
## Length:2571 Min. :5.040 Min. :23.63 Min. :0.07933
## Class :character 1st Qu.:5.293 1st Qu.:23.92 1st Qu.:0.23917
## Mode :character Median :5.347 Median :23.97 Median :0.27133
## Mean :5.370 Mean :23.97 Mean :0.27712
## 3rd Qu.:5.453 3rd Qu.:24.03 3rd Qu.:0.31200
## Max. :5.700 Max. :24.32 Max. :0.47800
## NA's :10 NA's :38 NA's :39
## Carb.Pressure Carb.Temp PSC PSC.Fill
## Min. :57.00 Min. :128.6 Min. :0.00200 Min. :0.0000
## 1st Qu.:65.60 1st Qu.:138.4 1st Qu.:0.04800 1st Qu.:0.1000
## Median :68.20 Median :140.8 Median :0.07600 Median :0.1800
## Mean :68.19 Mean :141.1 Mean :0.08457 Mean :0.1954
## 3rd Qu.:70.60 3rd Qu.:143.8 3rd Qu.:0.11200 3rd Qu.:0.2600
## Max. :79.40 Max. :154.0 Max. :0.27000 Max. :0.6200
## NA's :27 NA's :26 NA's :33 NA's :23
## PSC.CO2 Mnf.Flow Carb.Pressure1 Fill.Pressure
## Min. :0.00000 Min. : -100.20 Min. :105.6 Min. :34.60
## 1st Qu.:0.02000 1st Qu.: -100.00 1st Qu.:119.0 1st Qu.:46.00
## Median :0.04000 Median : 65.20 Median :123.2 Median :46.40
## Mean :0.05641 Mean : 24.57 Mean :122.6 Mean :47.92
## 3rd Qu.:0.08000 3rd Qu.: 140.80 3rd Qu.:125.4 3rd Qu.:50.00
## Max. :0.24000 Max. : 229.40 Max. :140.2 Max. :60.40
## NA's :39 NA's :2 NA's :32 NA's :22
## Hyd.Pressure2 Filler.Level Filler.Speed Temperature Usage.cont
## Min. : 0.00 Min. : 55.8 Min. : 998 Min. :63.60 Min. :12.08
## 1st Qu.: 0.00 1st Qu.: 98.3 1st Qu.:3888 1st Qu.:65.20 1st Qu.:18.36
## Median :28.60 Median :118.4 Median :3982 Median :65.60 Median :21.79
## Mean :20.96 Mean :109.3 Mean :3687 Mean :65.97 Mean :20.99
## 3rd Qu.:34.60 3rd Qu.:120.0 3rd Qu.:3998 3rd Qu.:66.40 3rd Qu.:23.75
## Max. :59.40 Max. :161.2 Max. :4030 Max. :76.20 Max. :25.90
## NA's :15 NA's :20 NA's :57 NA's :14 NA's :5
## Carb.Flow PH Oxygen.Filler Air.Pressurer
## Min. : 26 Min. :7.880 Min. :0.00240 Min. :140.8
## 1st Qu.:1144 1st Qu.:8.440 1st Qu.:0.02200 1st Qu.:142.2
## Median :3028 Median :8.540 Median :0.03340 Median :142.6
## Mean :2468 Mean :8.546 Mean :0.04684 Mean :142.8
## 3rd Qu.:3186 3rd Qu.:8.680 3rd Qu.:0.06000 3rd Qu.:143.0
## Max. :5104 Max. :9.360 Max. :0.40000 Max. :148.2
## NA's :2 NA's :4 NA's :12
```

After quickly browsing the remaining variables, while some of the fields have a wide range of values, none of the values look like errors (there are clusters of similar low/high values for each variable). While they may count as outliers (we'll calculate later) they are likely legitimate data and not input errors.

Let's check histograms to see if anything stands out:

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



```
##   Brand.Code    n
## 1         A    293
## 2         B   1239
## 3         C    304
## 4         D    615
## 5        <NA>   120
```

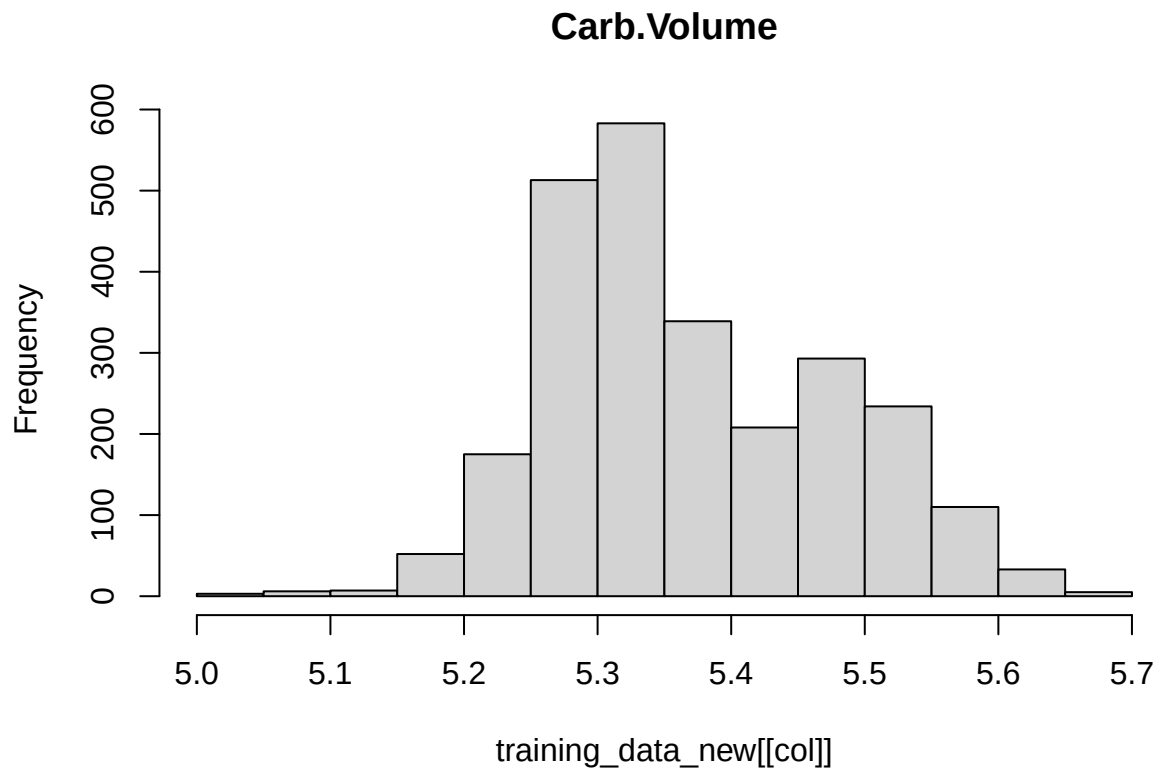
Not all the data is normally distributed. pH, the response variable is normally distributed. Other variables close to a normal distribution include carb pressure, carb pressure 1 (slightly bimodal), carb temp, fill ounces, carb volume (also slightly bimodal), psc (skewed slightly right), pc volume, temperature (skewed slightly right), and psc fill (skewed right with a large peak).

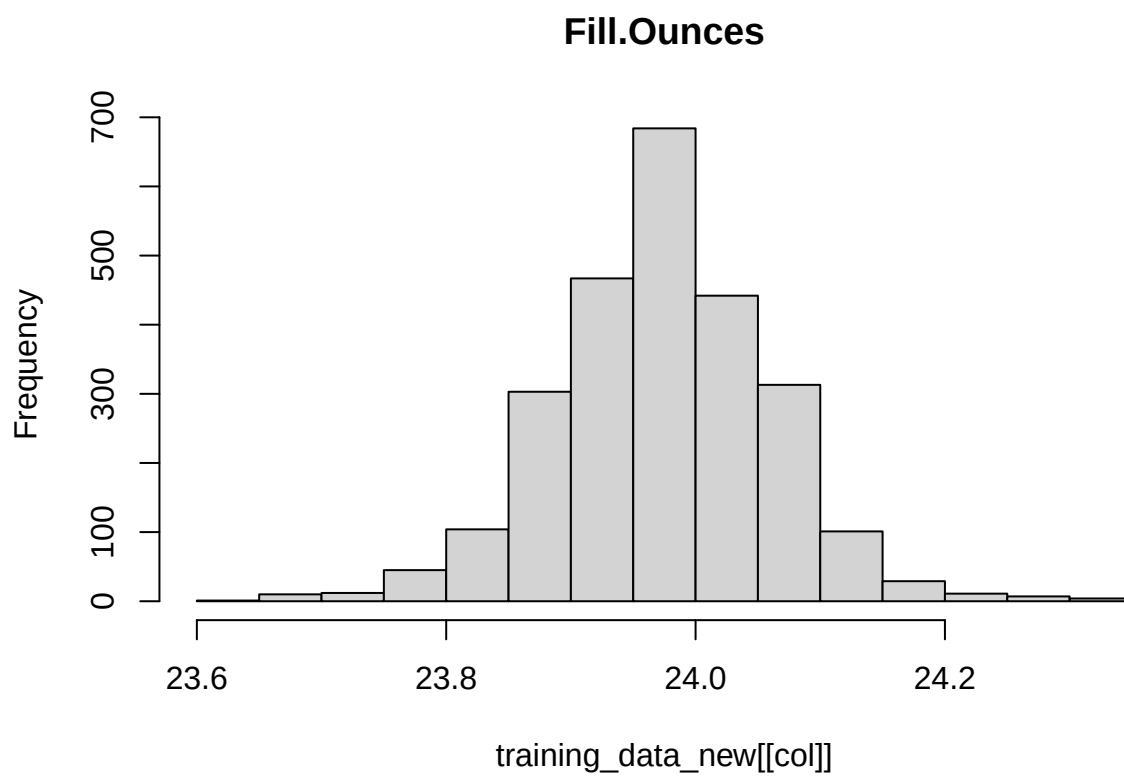
A few are more clearly bimodal/have very different peaks on either side of the graph: air pressurer, carb flow, fill pressure, filler level, filler speed, , hyd pressure 2, mnf flow, and usage cont.

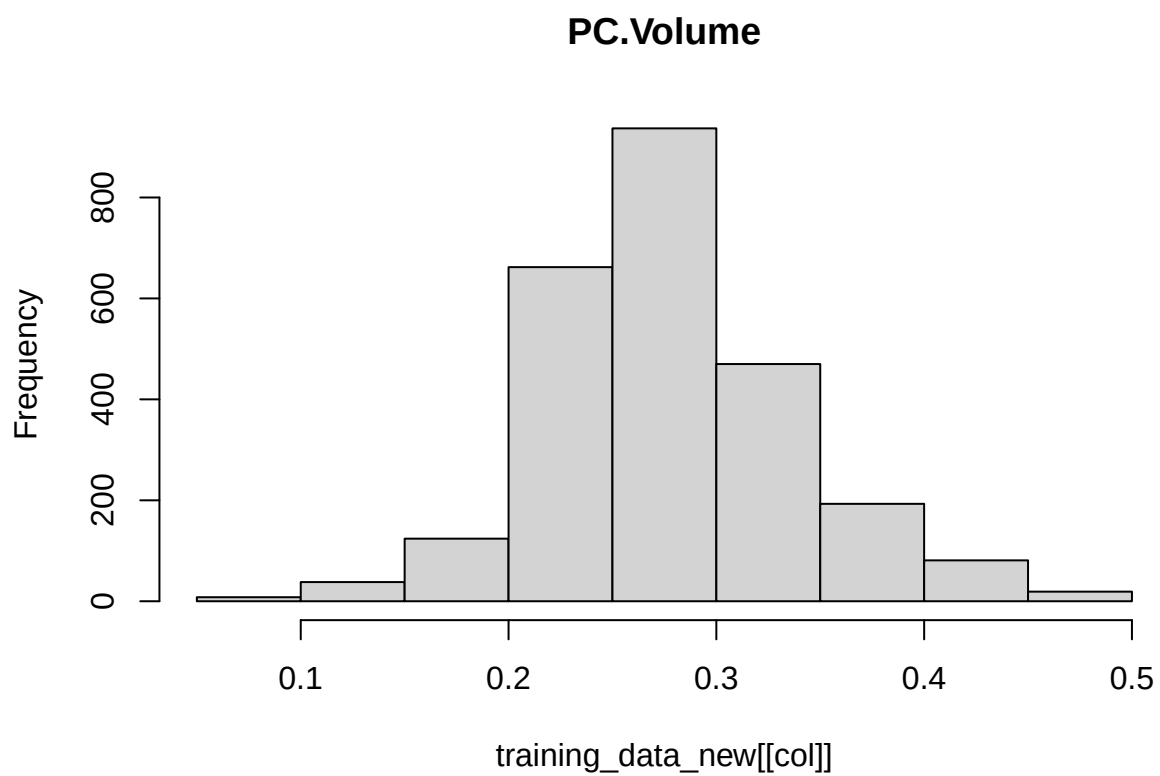
Finally, some are skewed right: in addition to the variables mentioned above, oxygen filler looks like it may have some outliers, and psc co2 has a strange distribution, where the variable clusters around certain values, almost like a discrete variable.

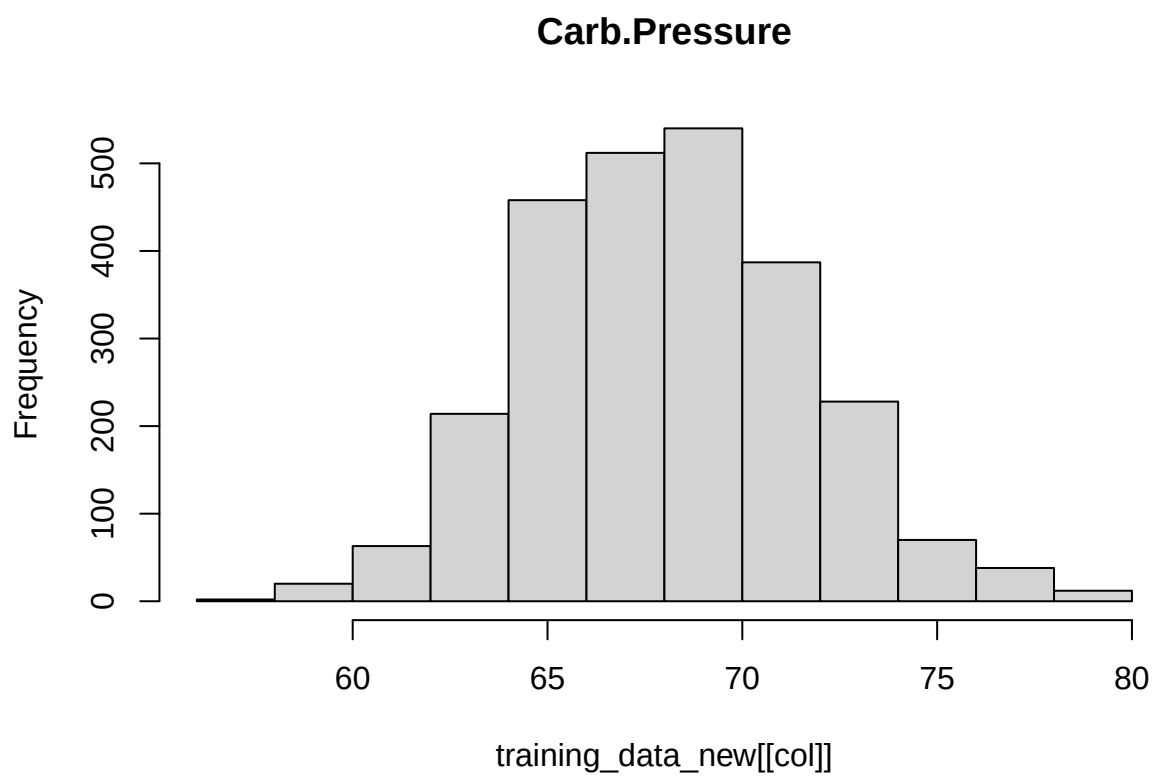
```
#histograms
```

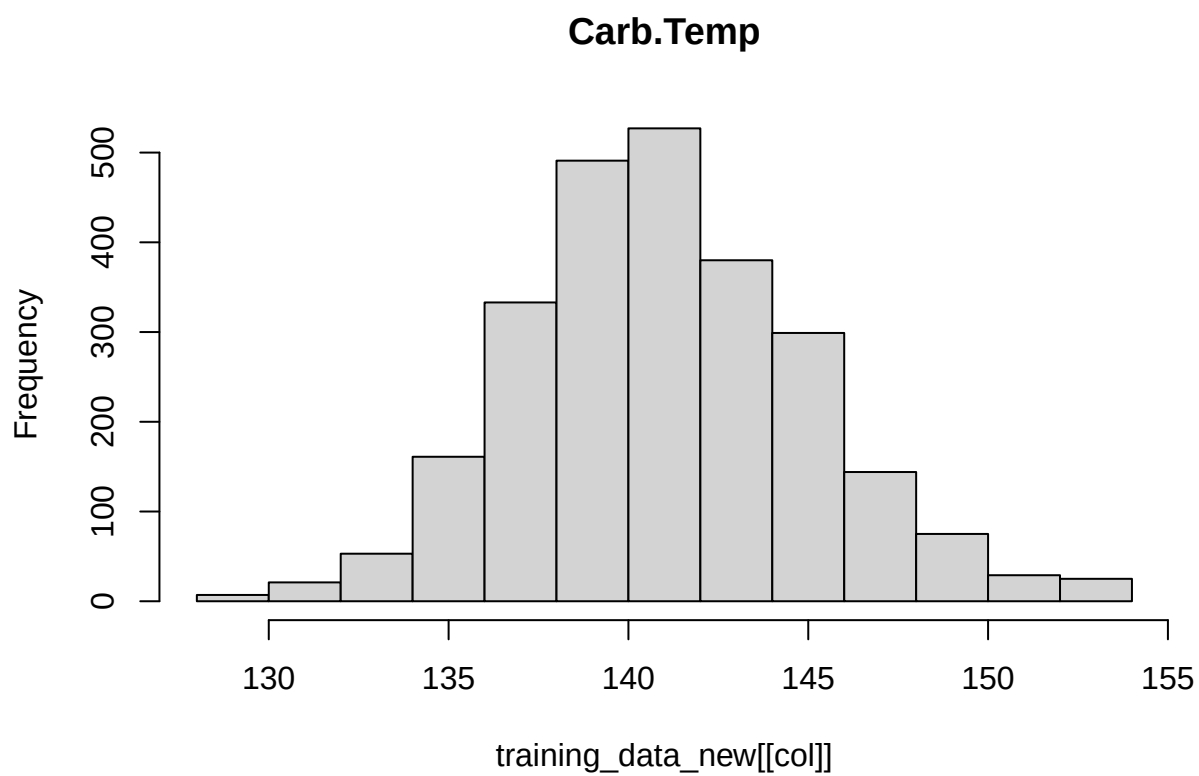
```
for (col in names(training_data_new %>% select(-Brand.Code))) {  
  hist(training_data_new[[col]], main=col)  
}
```



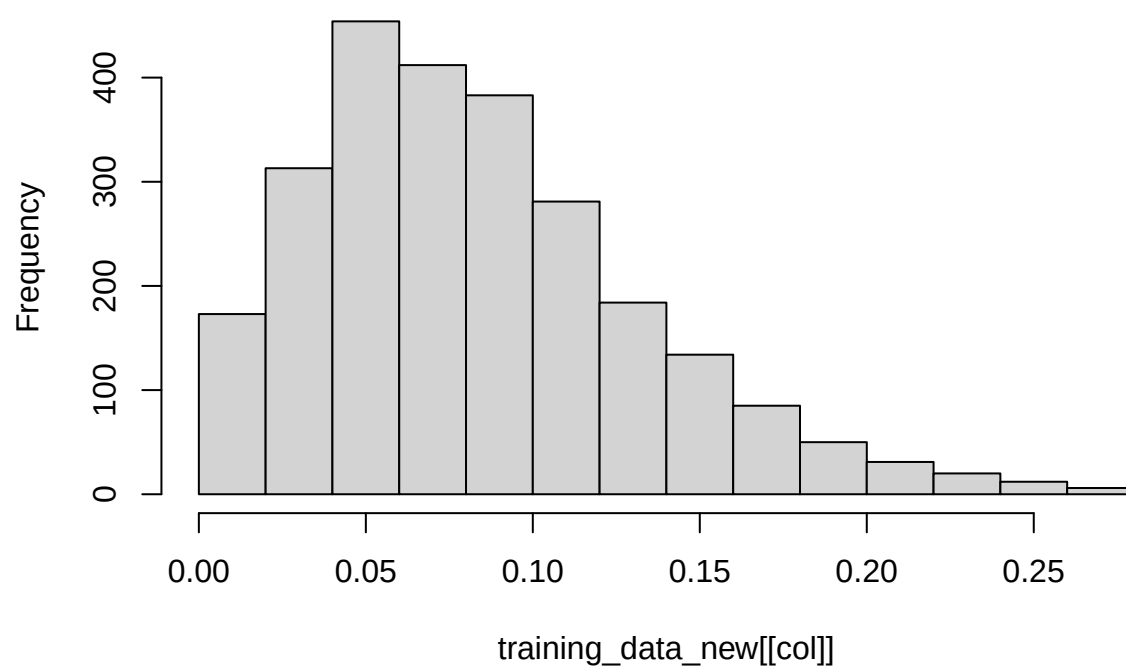




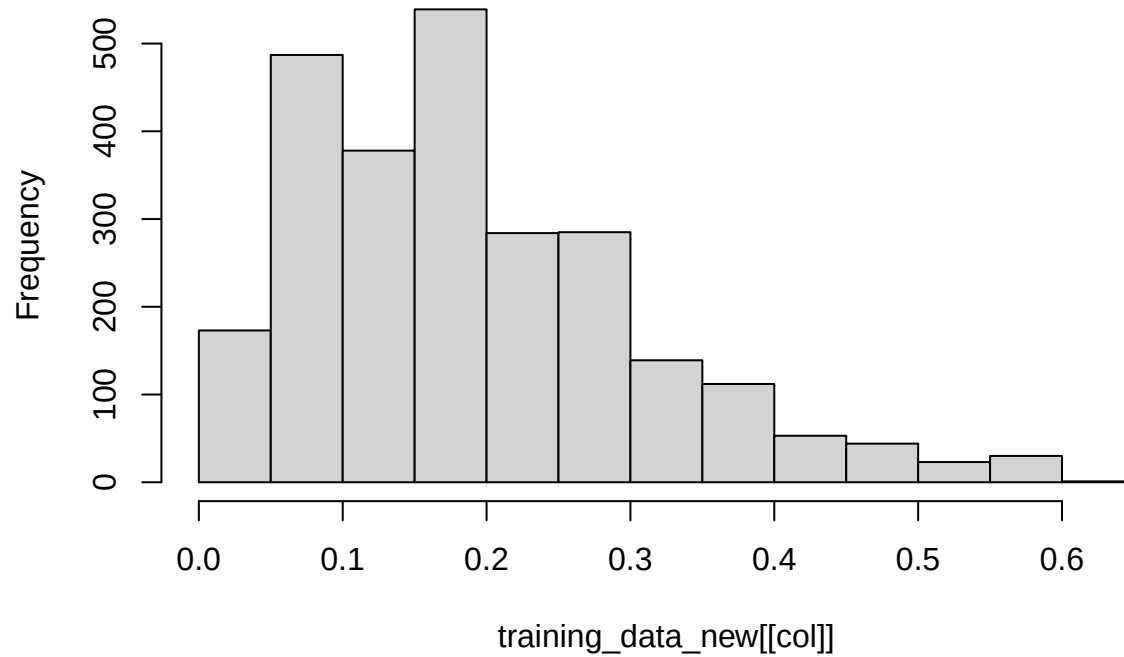




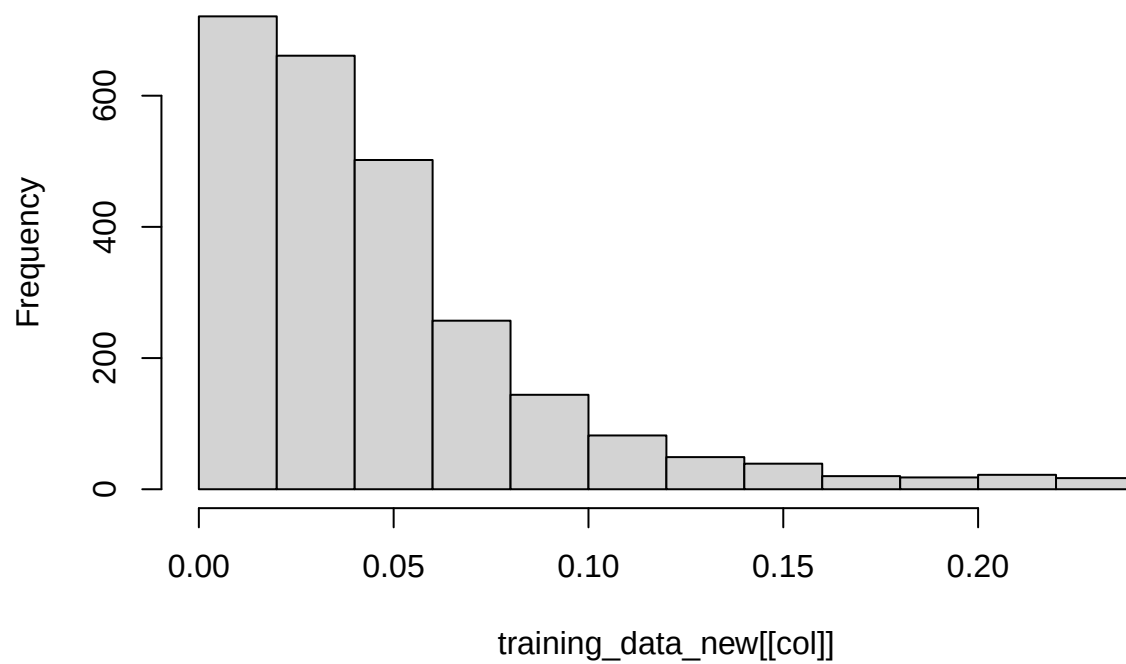
PSC



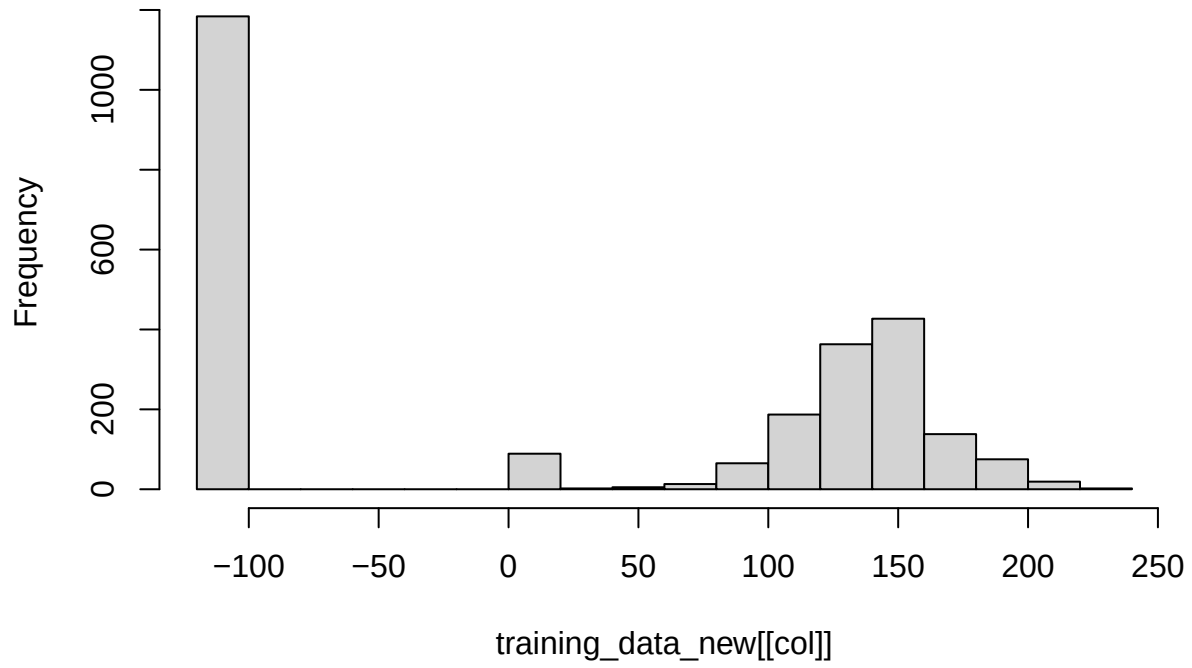
PSC.Fill

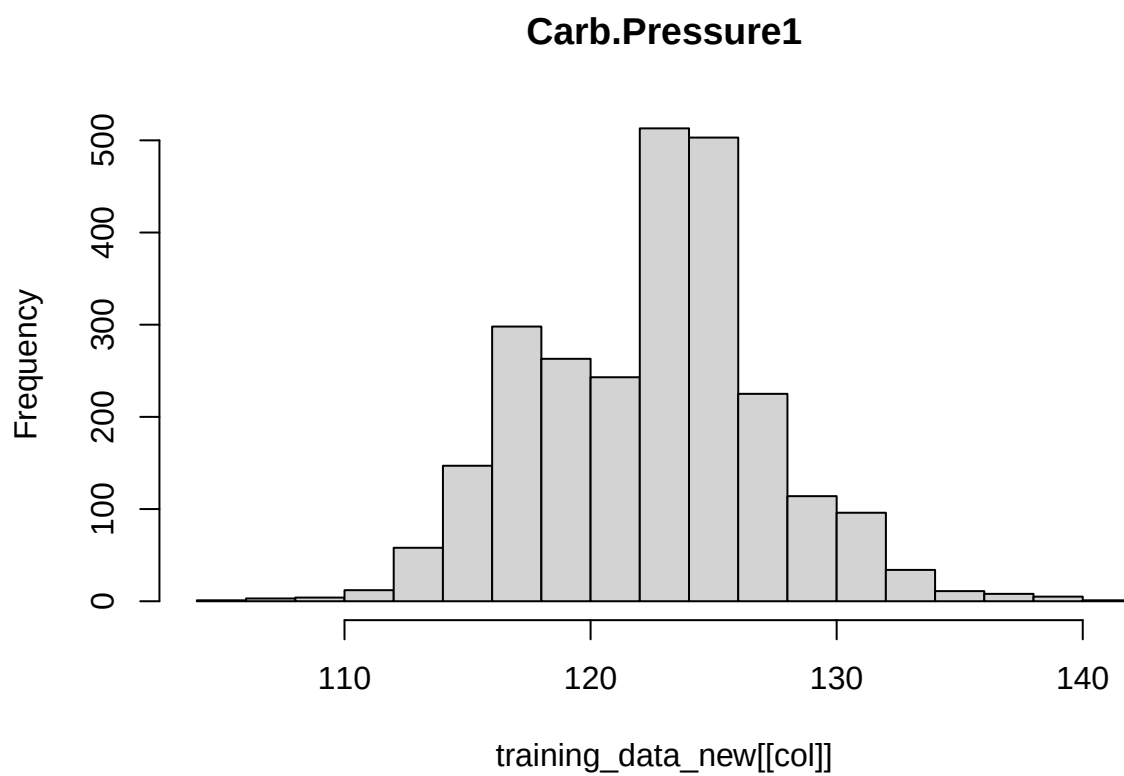


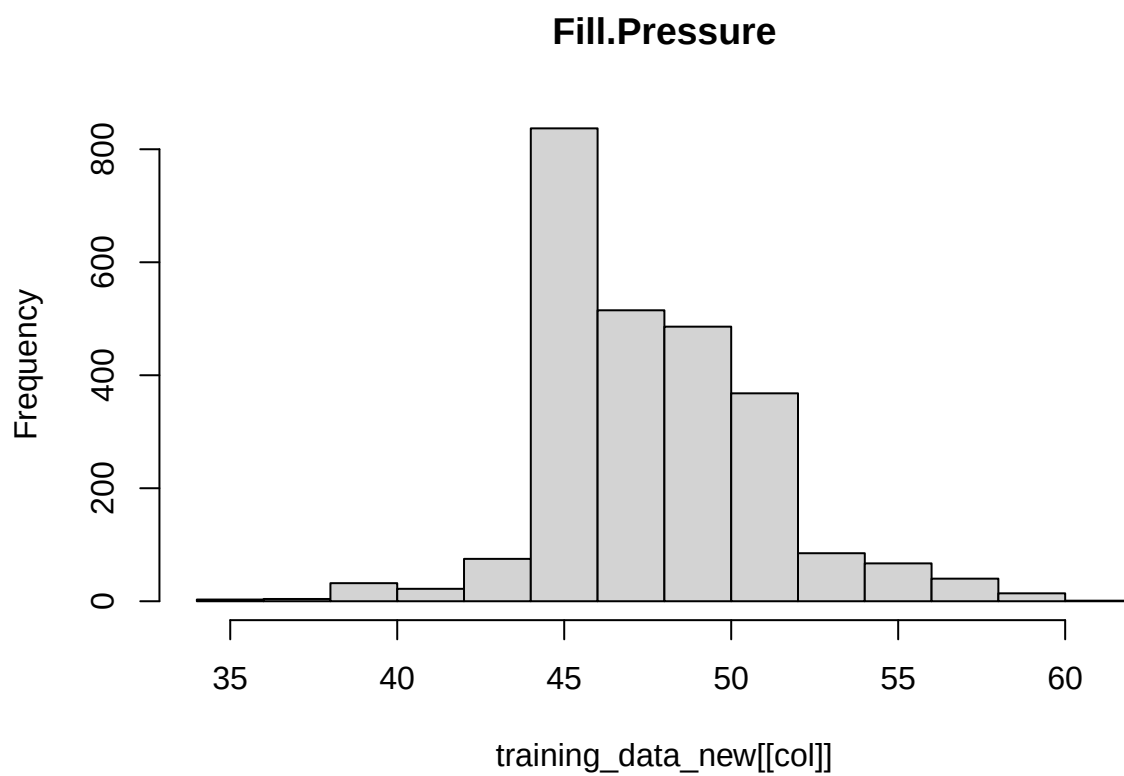
PSC.CO2



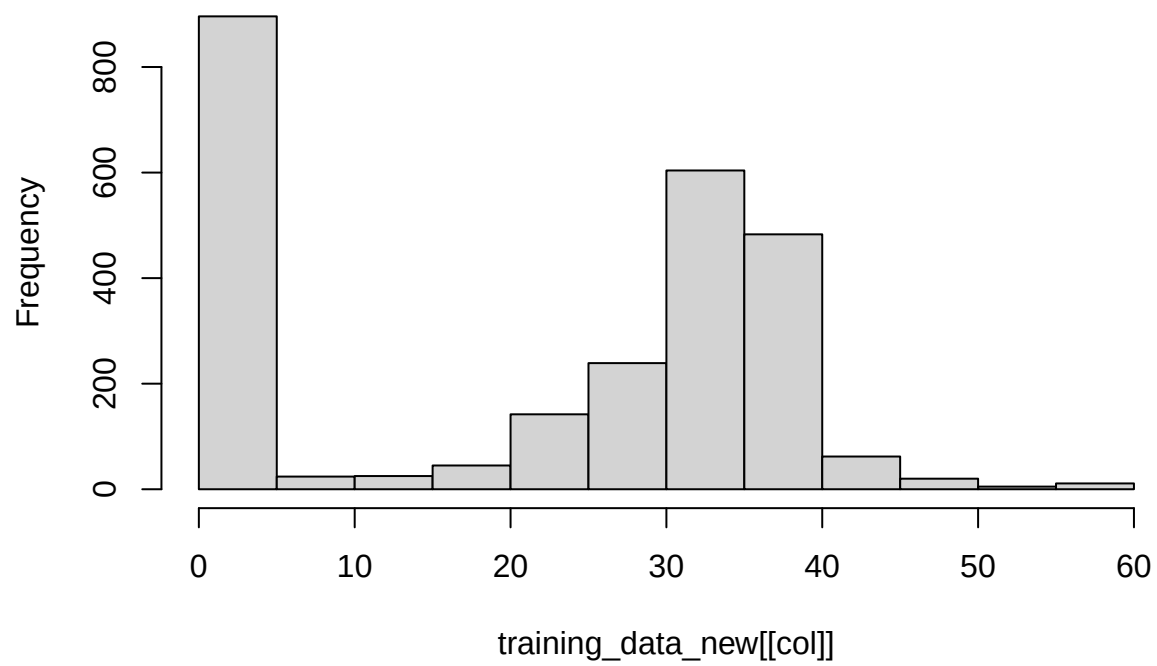
Mnf.Flow

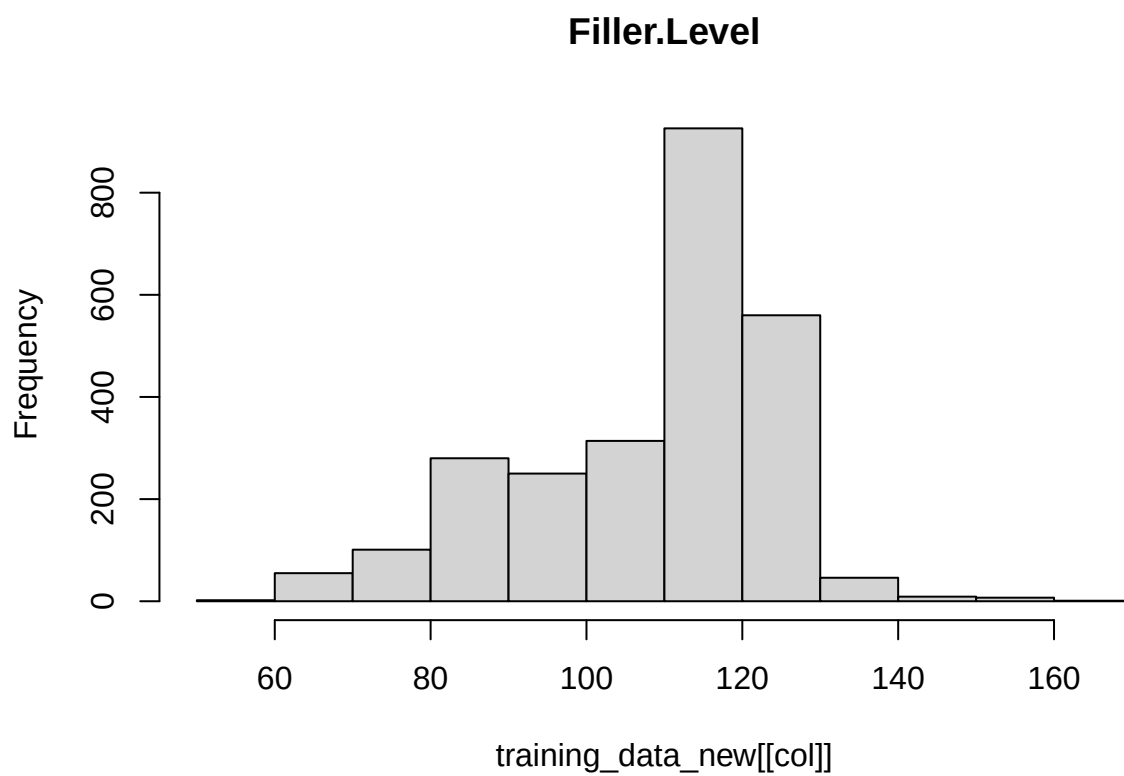


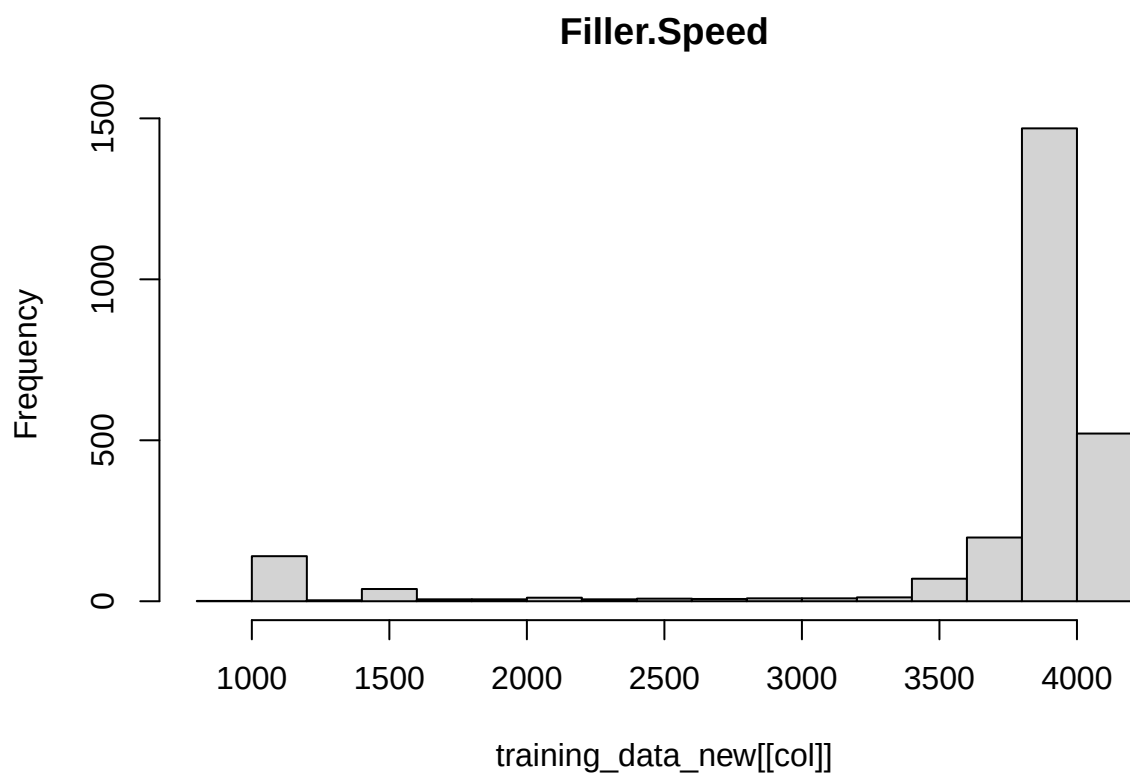


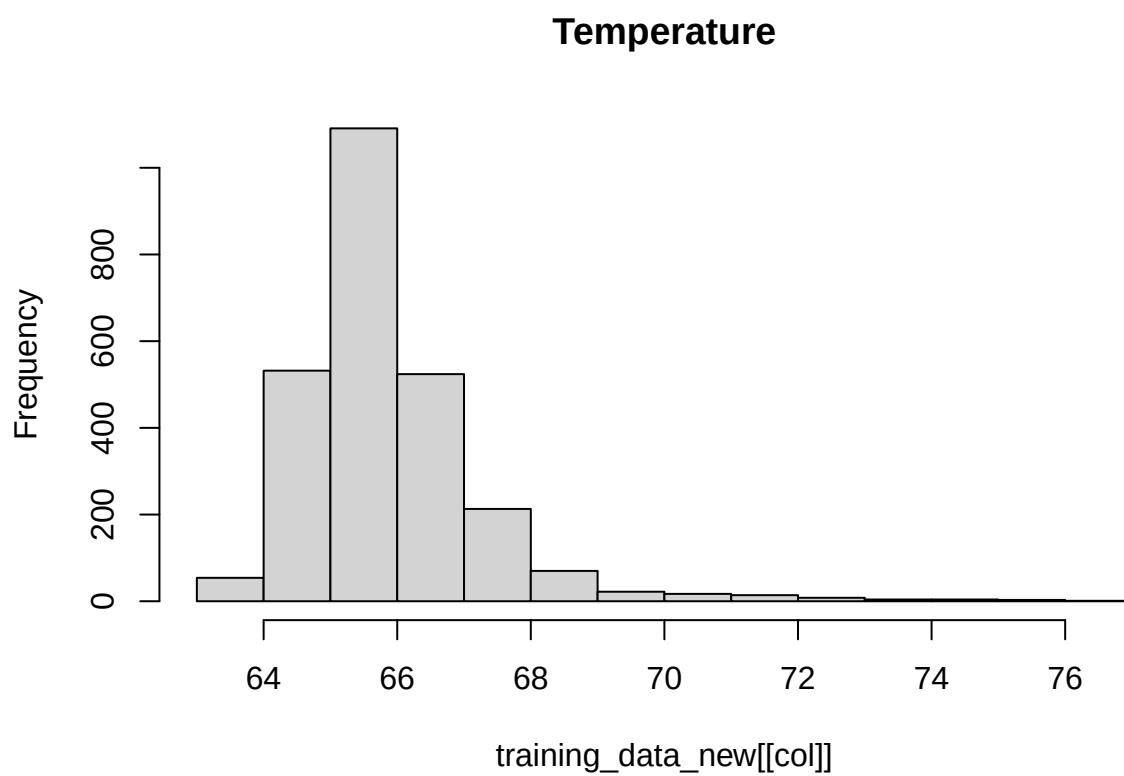


Hyd.Pressure2

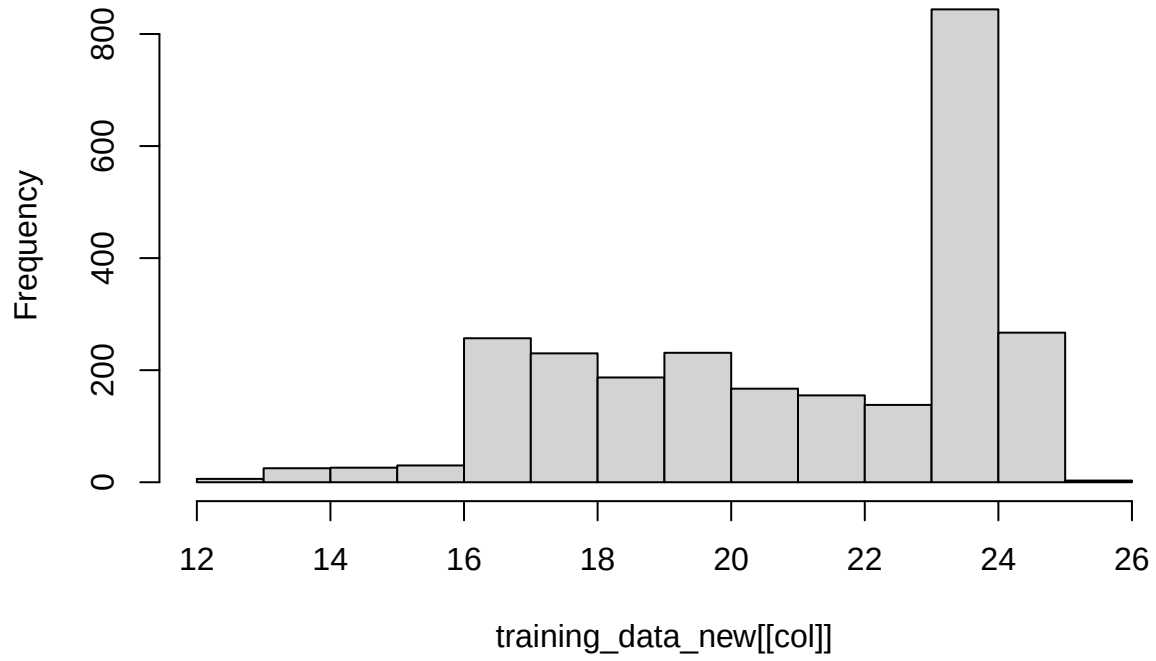


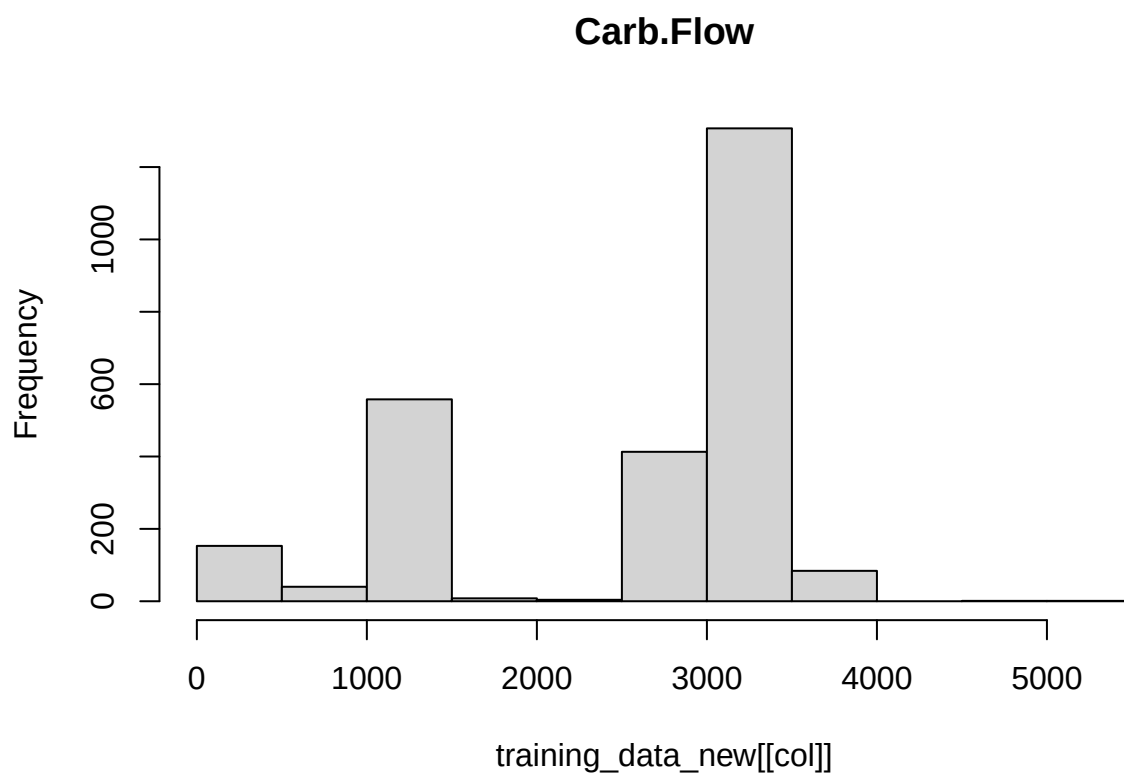


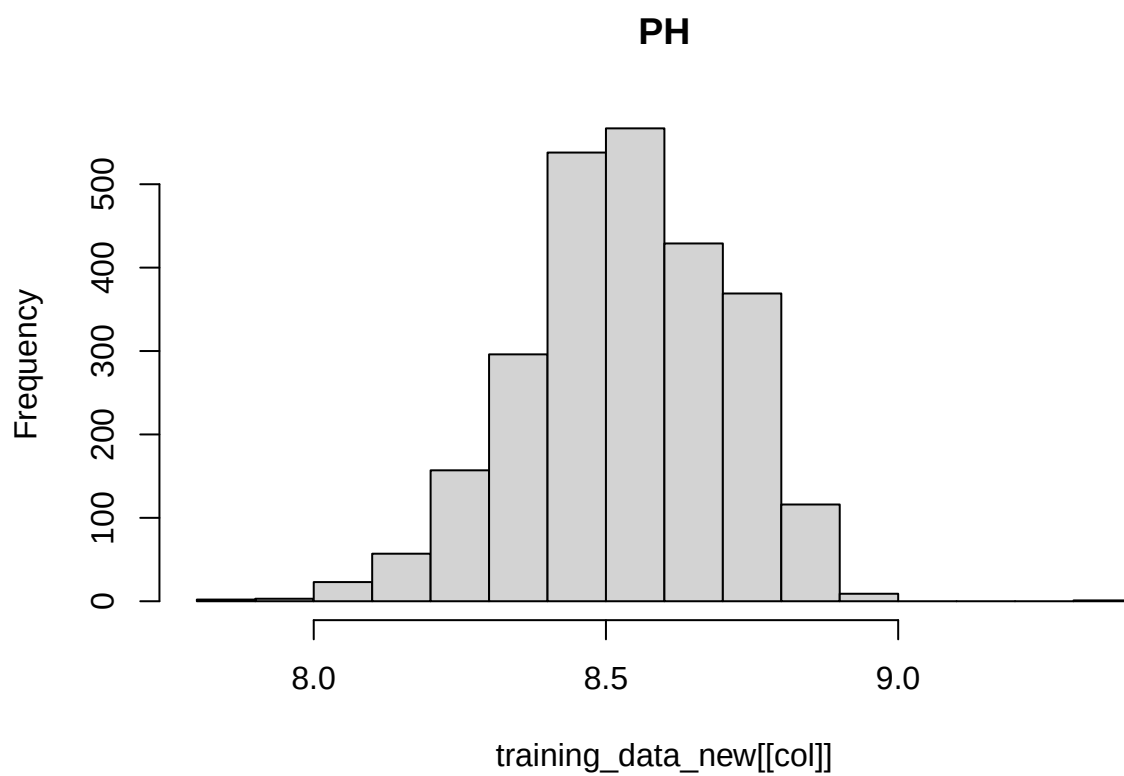




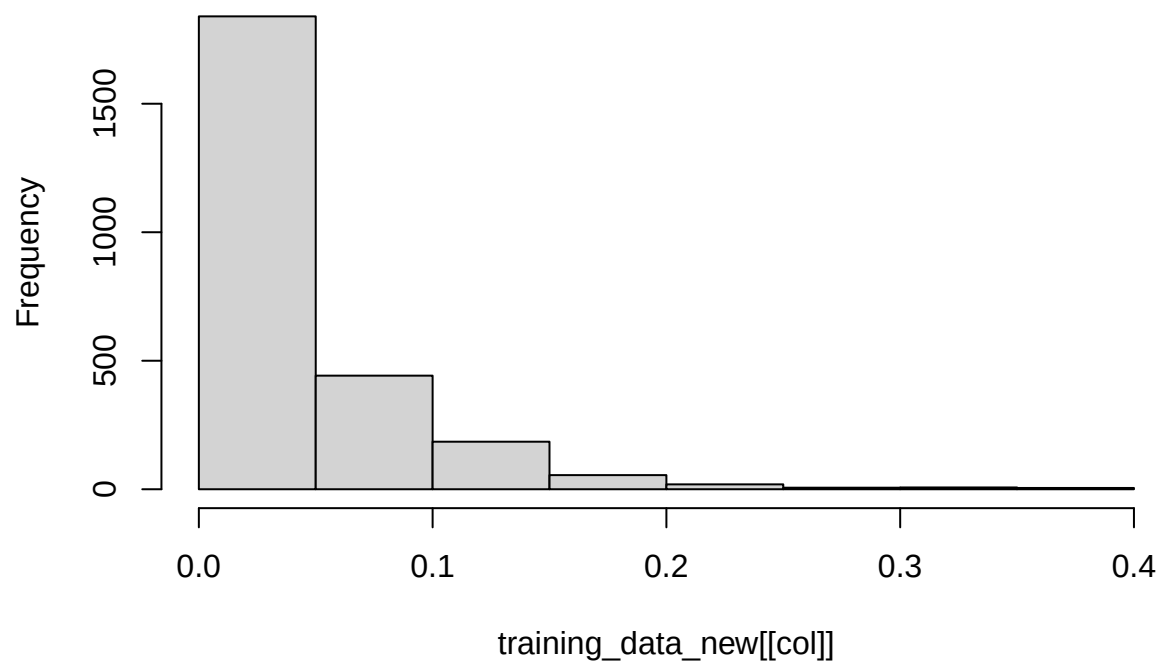
Usage.cont



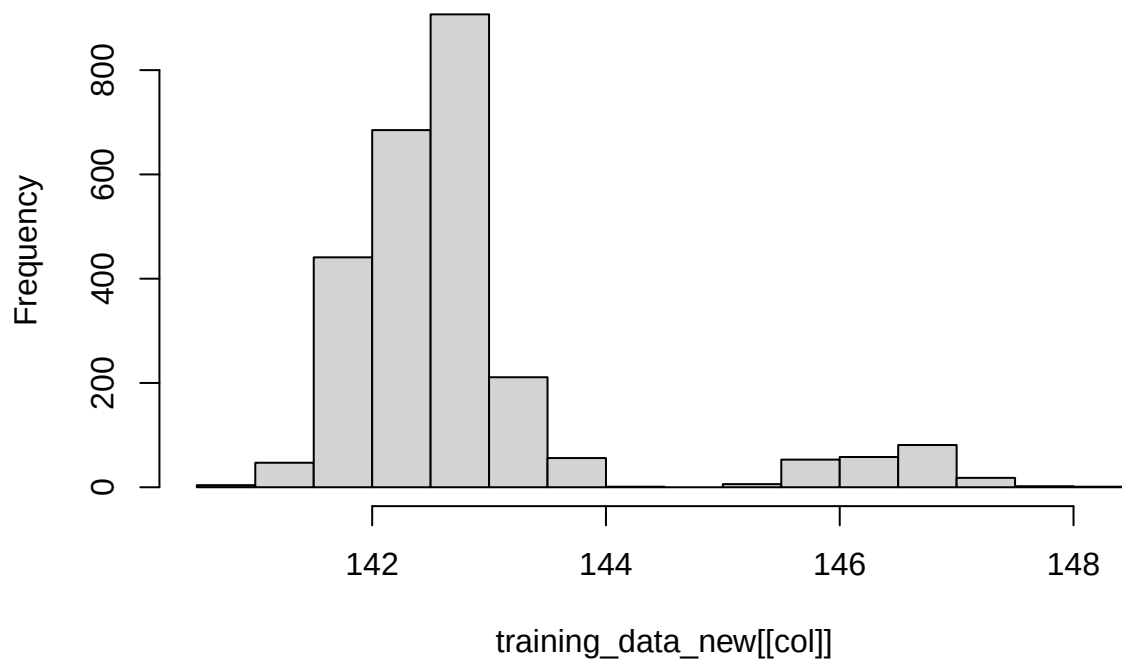




Oxygen.Filler

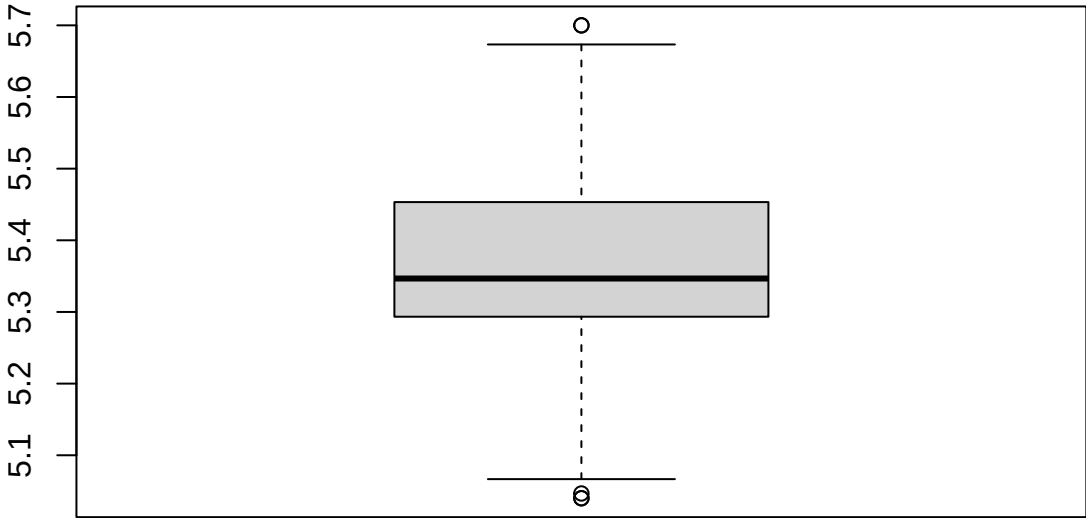


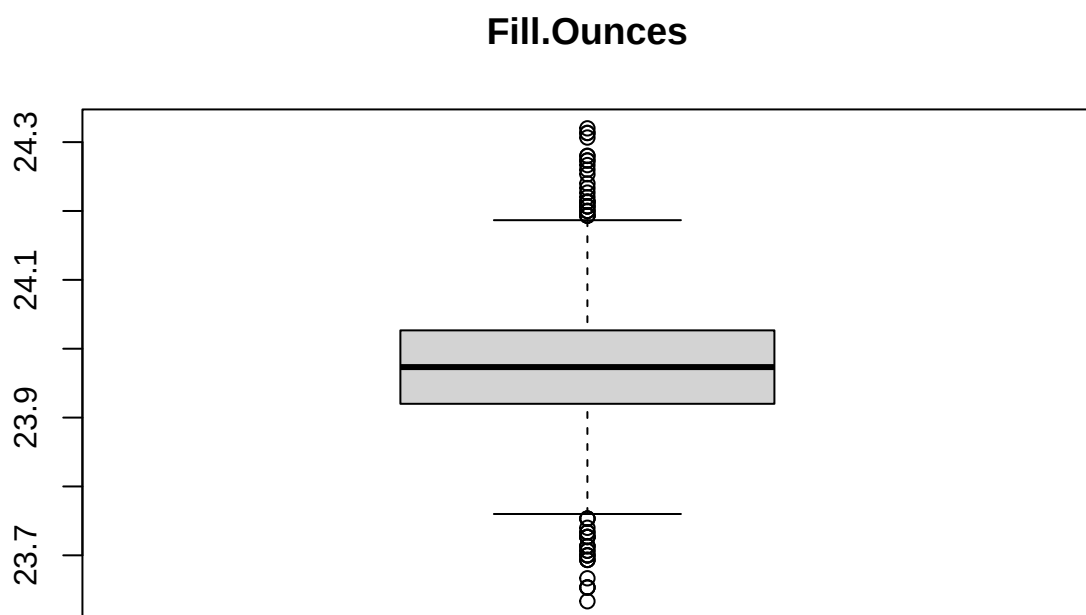
Air.Pressurer



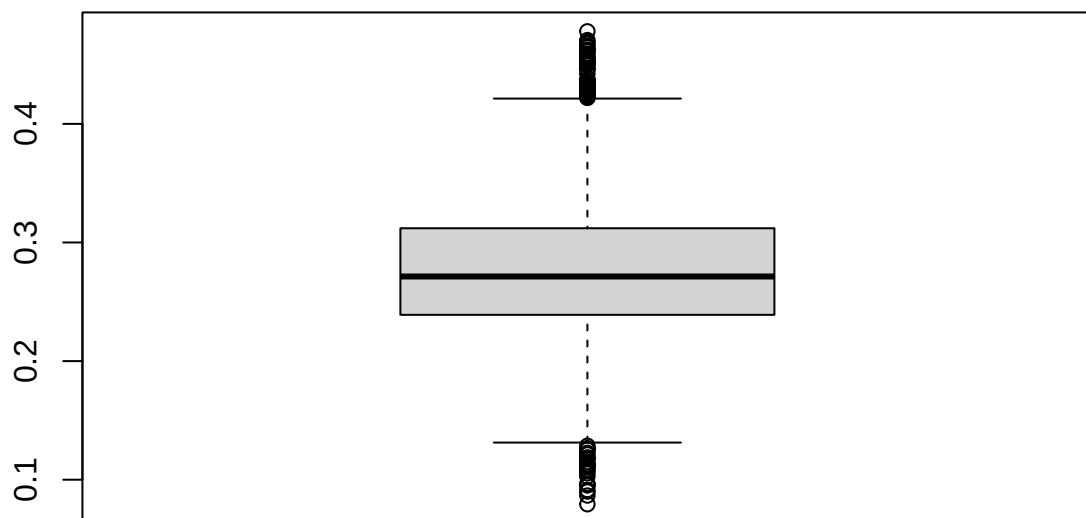
```
#box plots
for (col in names(training_data_new %>% select(-Brand.Code))) {
  boxplot(training_data_new[[col]], main=col)
}
```

Carb.Volume

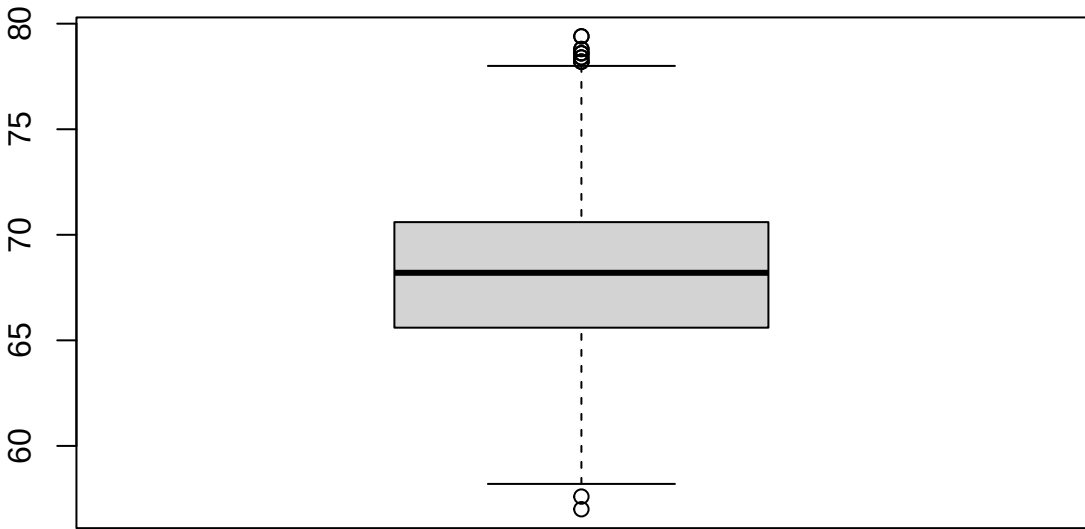


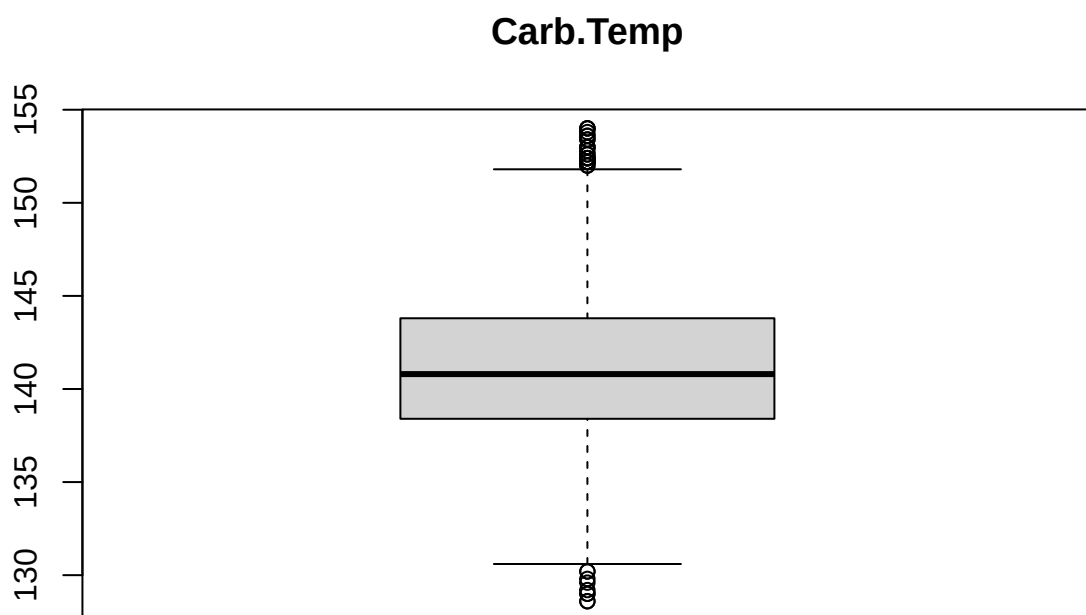


PC.Volume

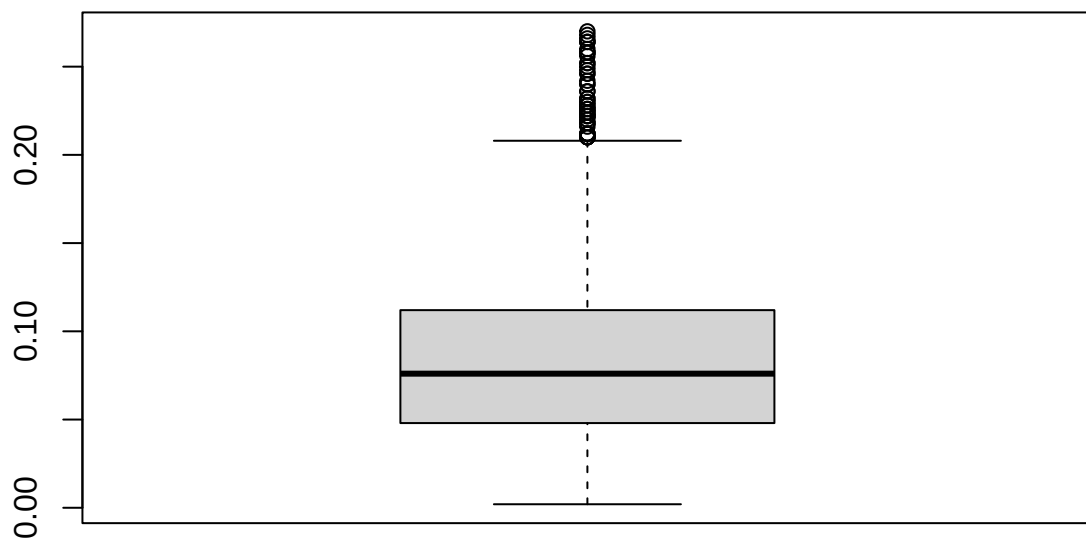


Carb.Pressure

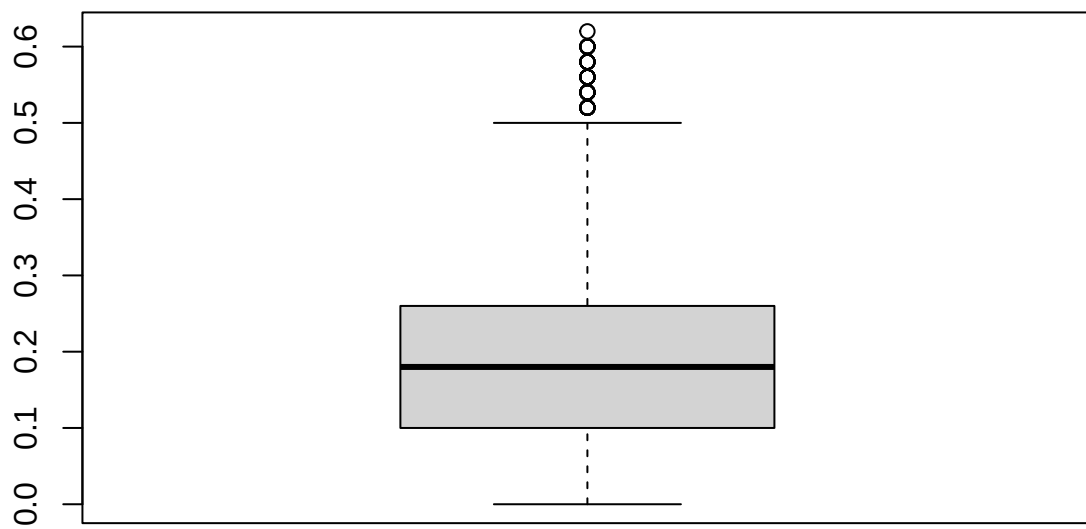




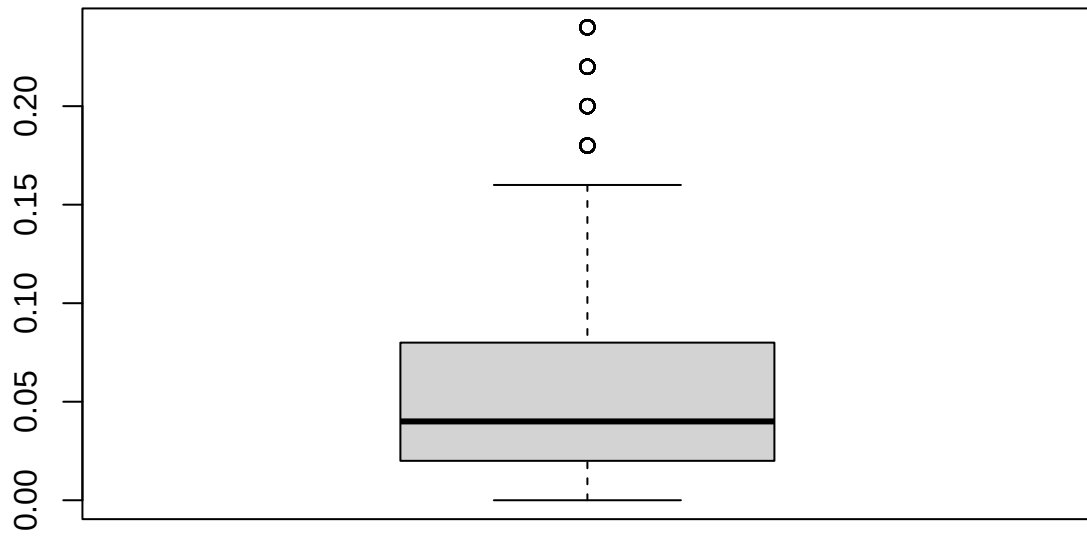
PSC



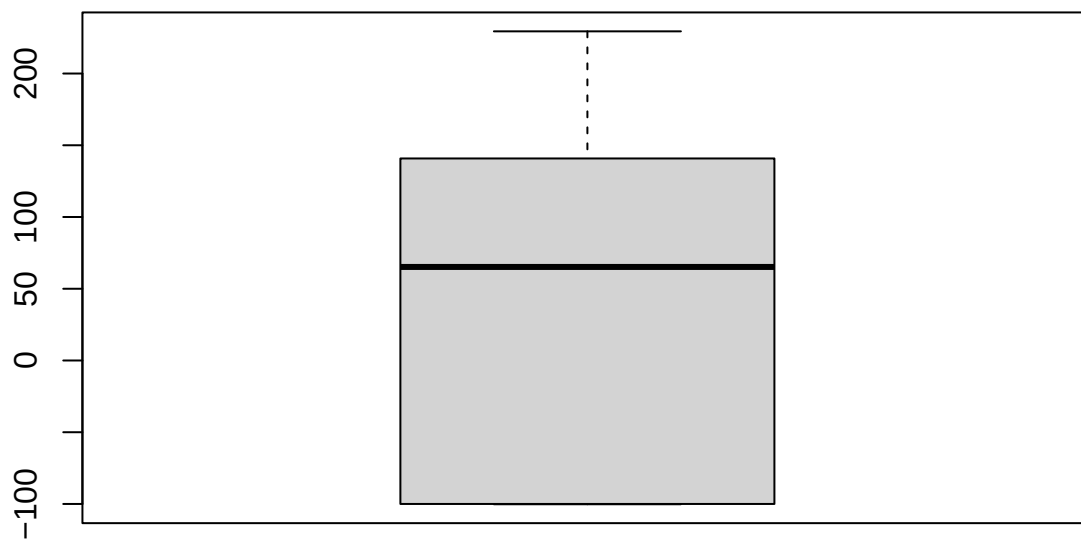
PSC.Fill



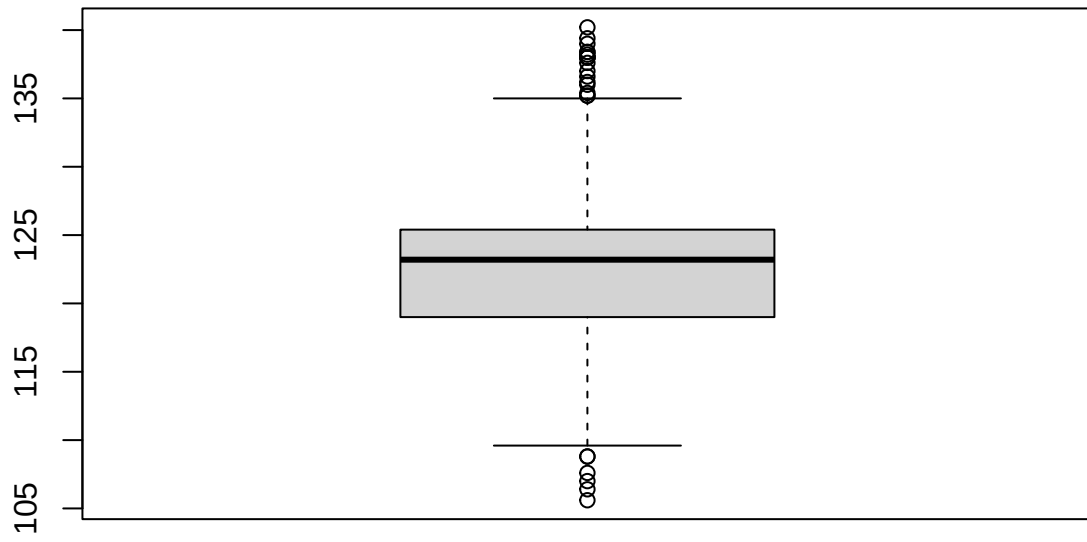
PSC.CO2



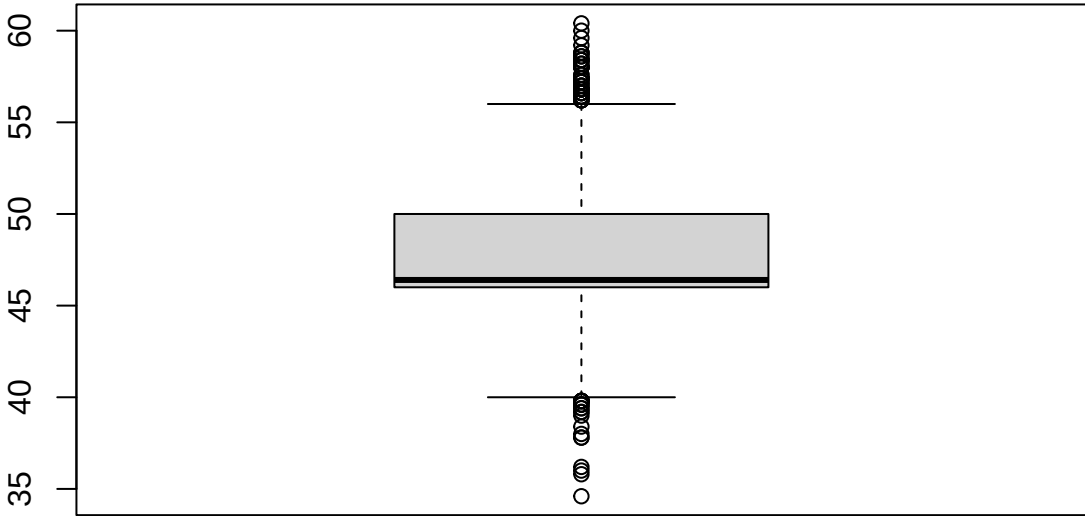
Mnf.Flow



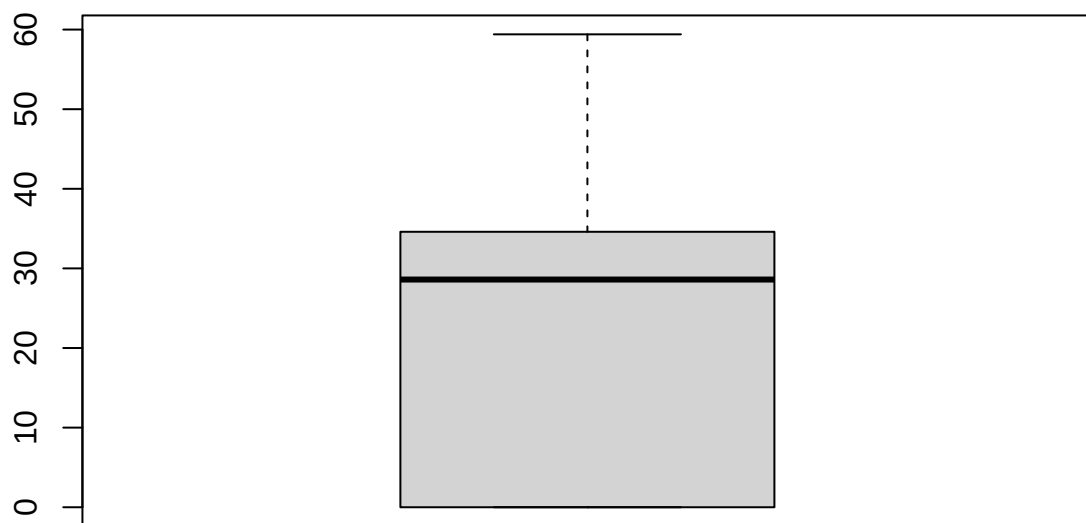
Carb.Pressure1

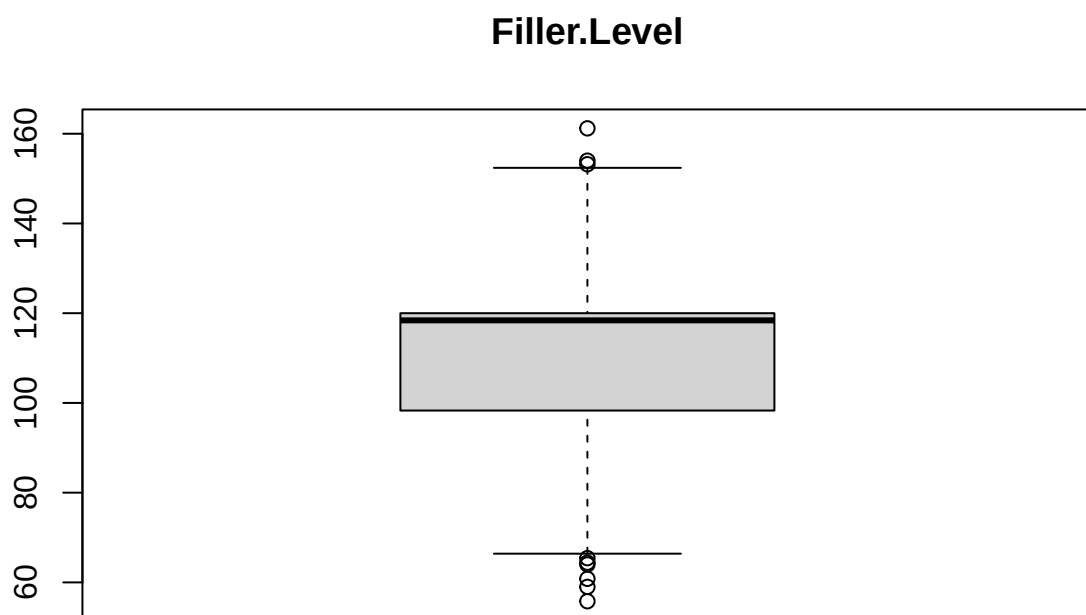


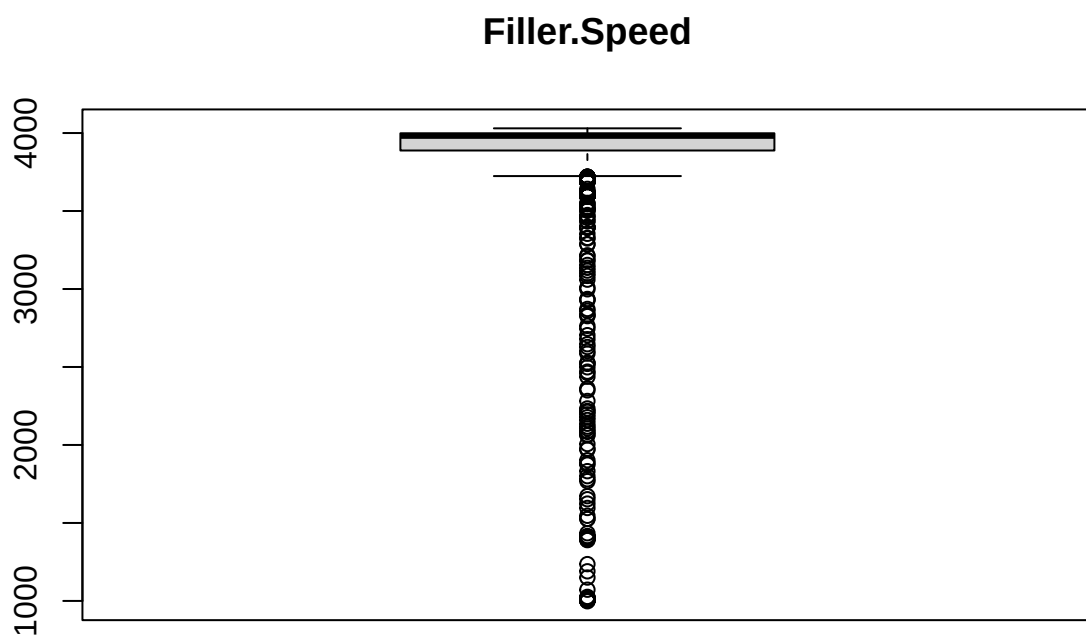
Fill.Pressure



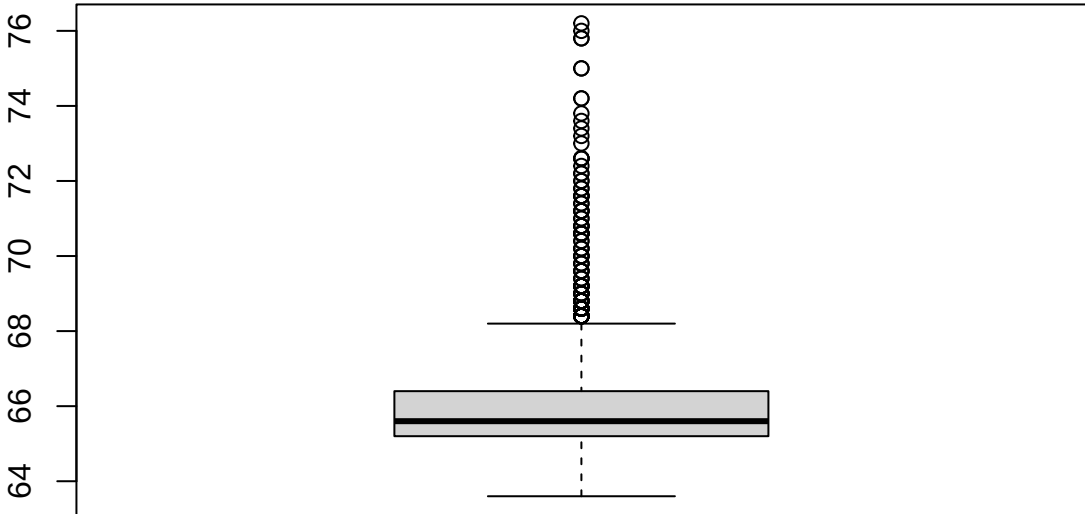
Hyd.Pressure2



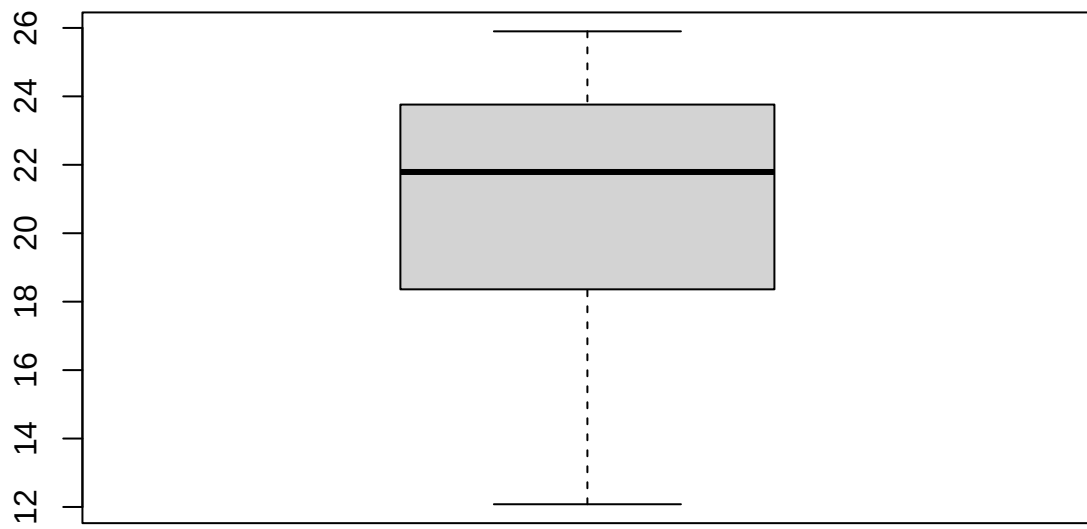




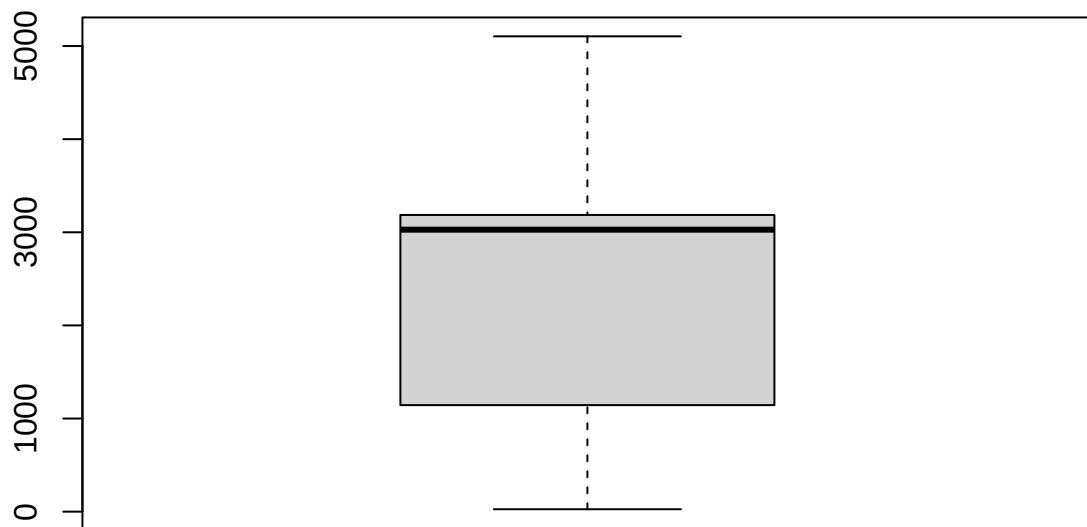
Temperature

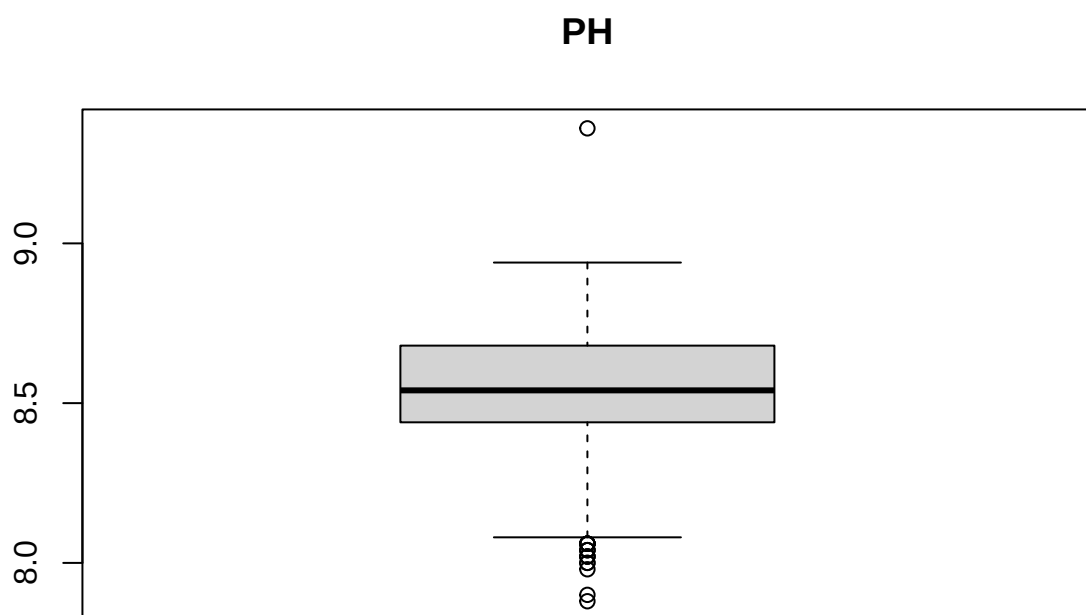


Usage.cont

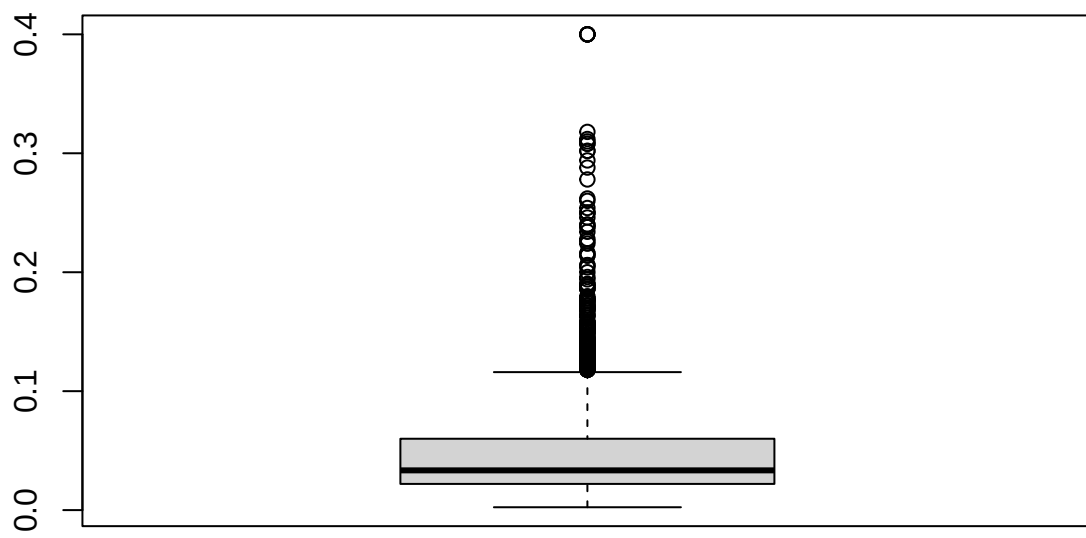


Carb.Flow

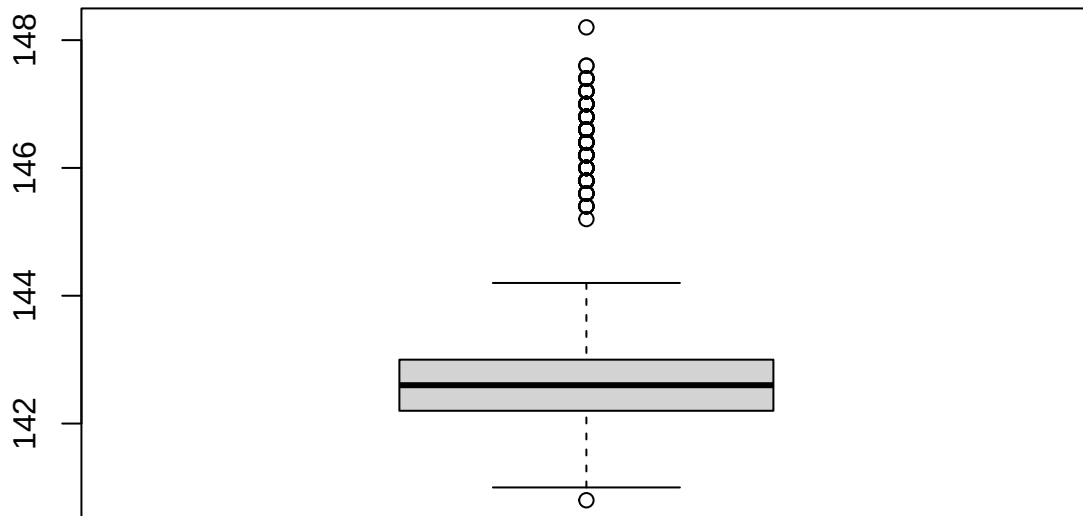




Oxygen.Filler



Air.Pressurer



Calculating the number of outliers in each column based on the IQR method

```
outliers <- sapply(training_data_new |> select(-Brand.Code), function(x) {
  Q1 <- quantile(x, 0.25, na.rm=TRUE)
  Q3 <- quantile(x, 0.75, na.rm=TRUE)
  IQR <- Q3 - Q1
  sum(x < Q1 - 1.5*IQR | x > Q3 + 1.5*IQR, na.rm=TRUE)
})
outliers
```

##	Carb.Volume	Fill.Ounces	PC.Volume	Carb.Pressure	Carb.Temp
##	5	55	86	14	36
##	PSC	PSC.Fill	PSC.CO2	Mnf.Flow	Carb.Pressure1
##	52	54	77	0	24
##	Fill.Pressure	Hyd.Pressure2	Filler.Level	Filler.Speed	Temperature
##	79	0	9	439	123
##	Usage.cont	Carb.Flow	PH	Oxygen.Filler	Air.Pressurer
##	0	0	18	189	220

Cross-checking the graphs with the data set, we could remove the one Filler.Level value that appears on the top of the box plot, 161.2 when the next-highest is 154. pH has one comparatively high value (9.36) that's nowhere close to the next-highest value. We can remove that row - it's missing 4 other variables. Other outlier values are one of several similar values, and could be categorized that way due to the shape of the distribution and wide range of values. For example, it looks like there's one particularly high value for Oxygen.Filler, but when we check there are 5 cases with the same value.

```

#removing the one outliers
training_data_new <- training_data_new |>
  filter(PH < 9 | is.na(PH)) |>
  filter(Filler.Level < 160 | is.na(Filler.Level))

#removing pH NAs because we shouldn't impute the response variable
training_data_no_na <- training_data_new |>
  filter(!is.na(PH))

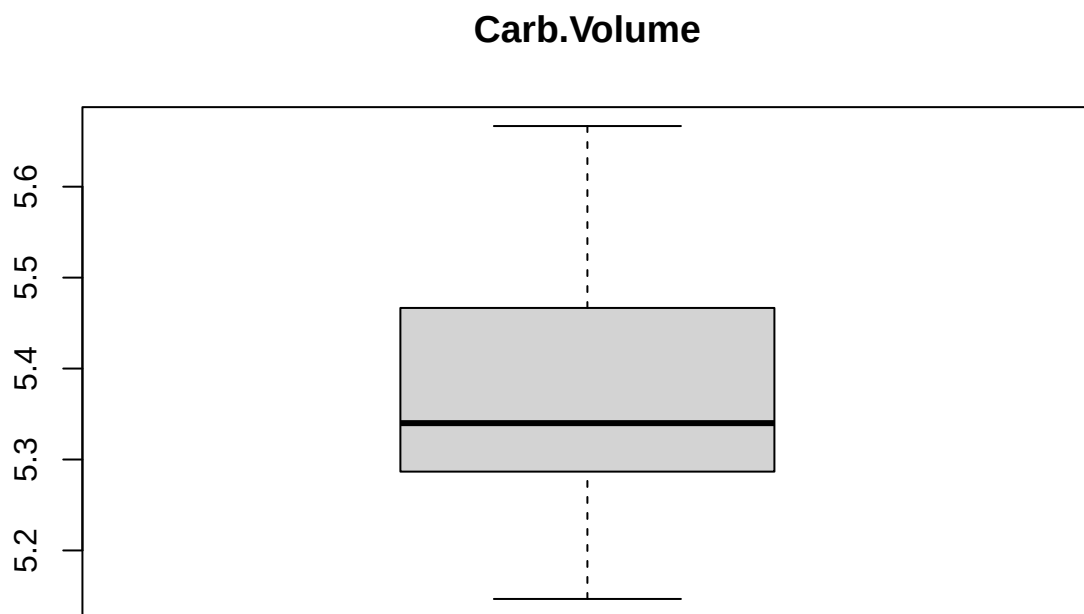
```

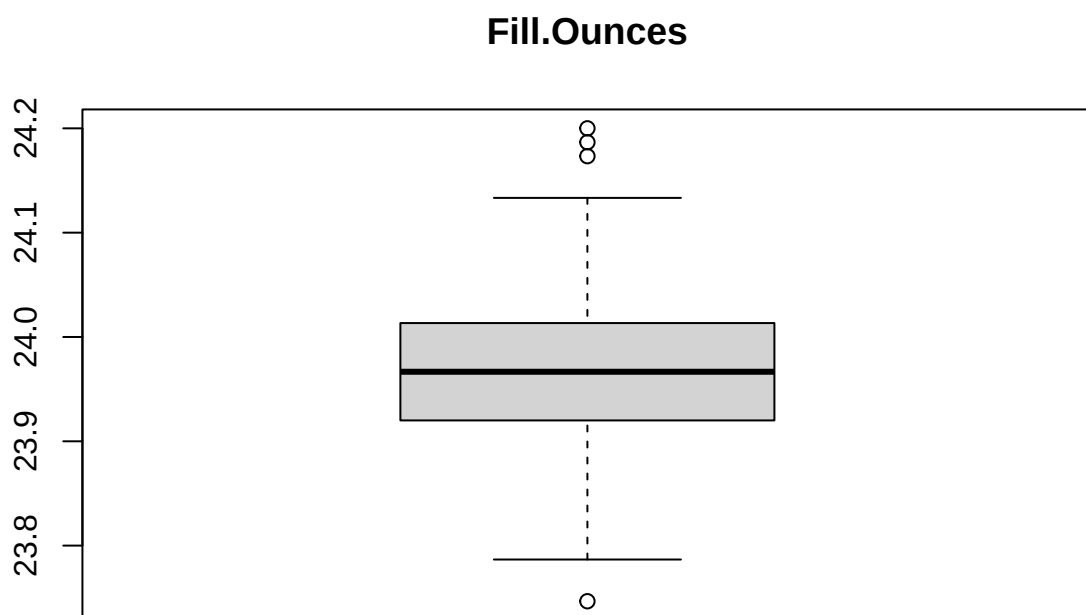
Looking at outliers in the test data set

```

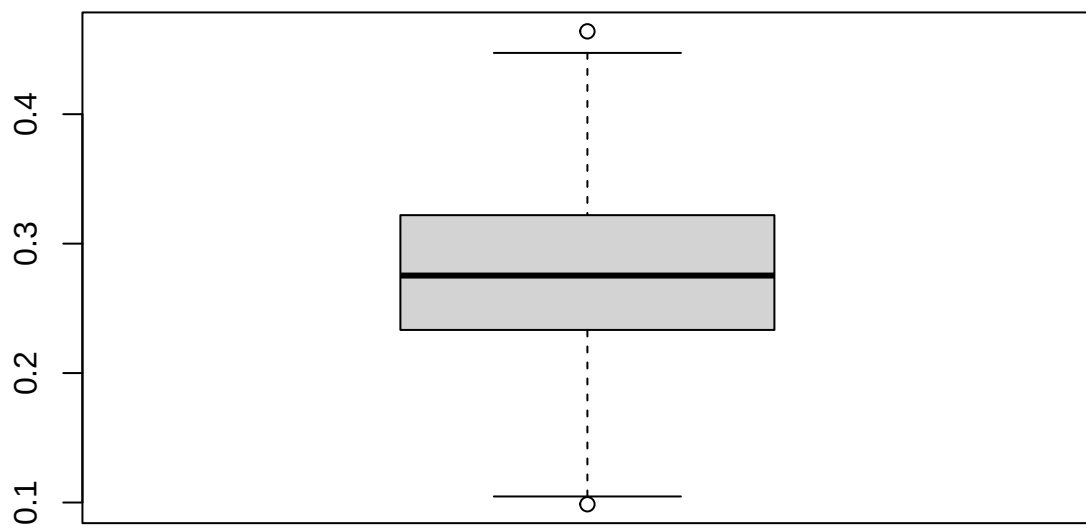
#leaving out the categorical and blank response variable
for (col in names(test_data_new %>% select(-Brand.Code, -PH))) {
  boxplot(test_data_new[[col]], main=col)
}

```

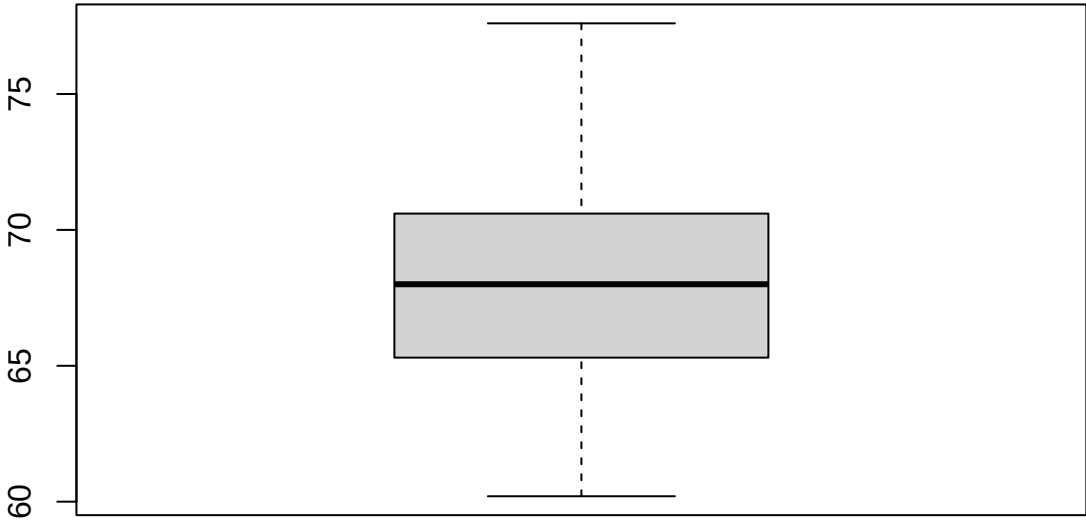




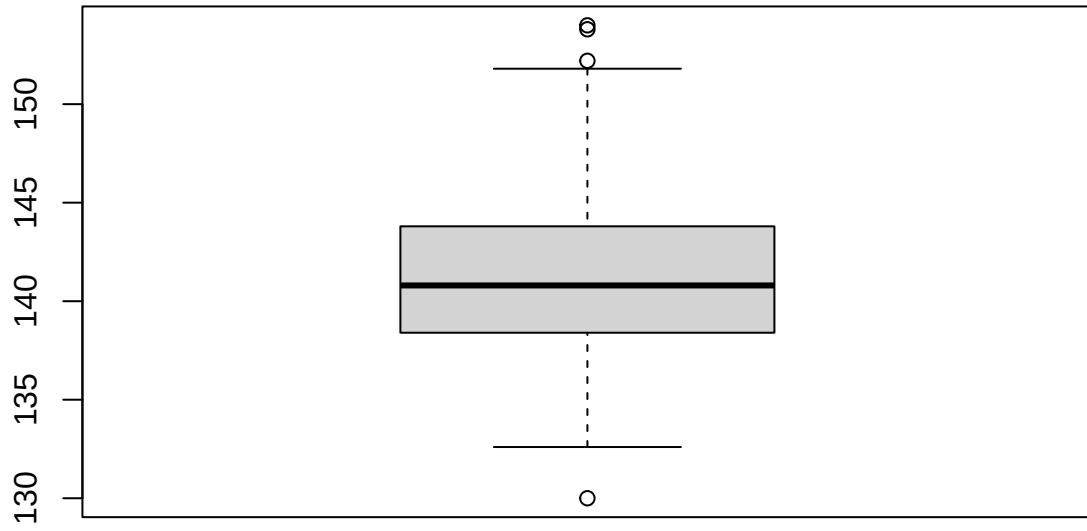
PC.Volume

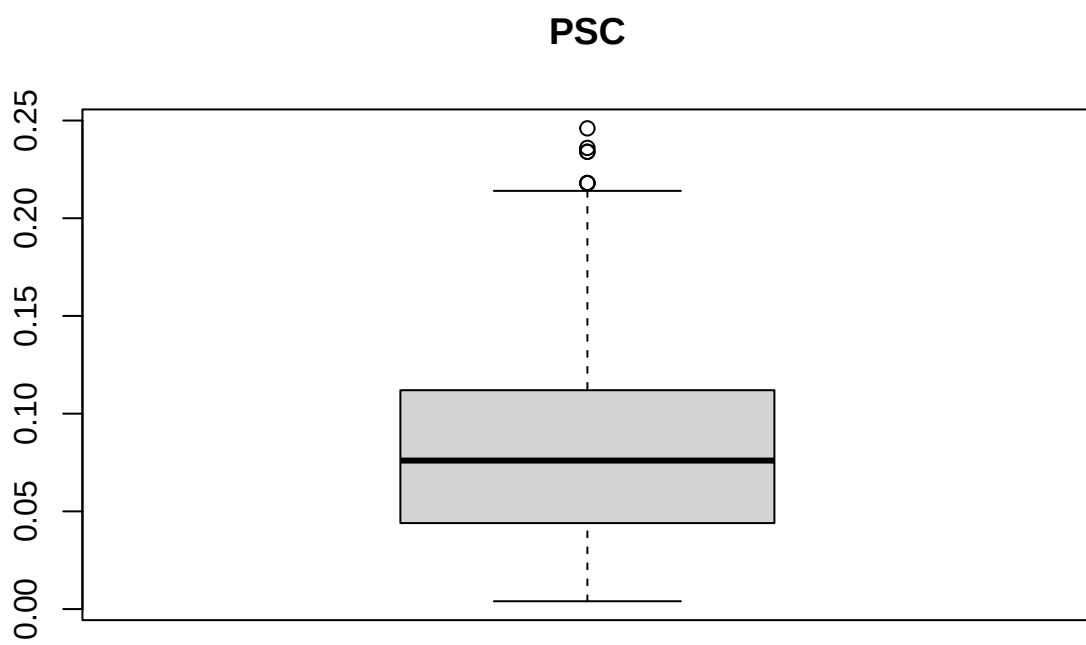


Carb.Pressure

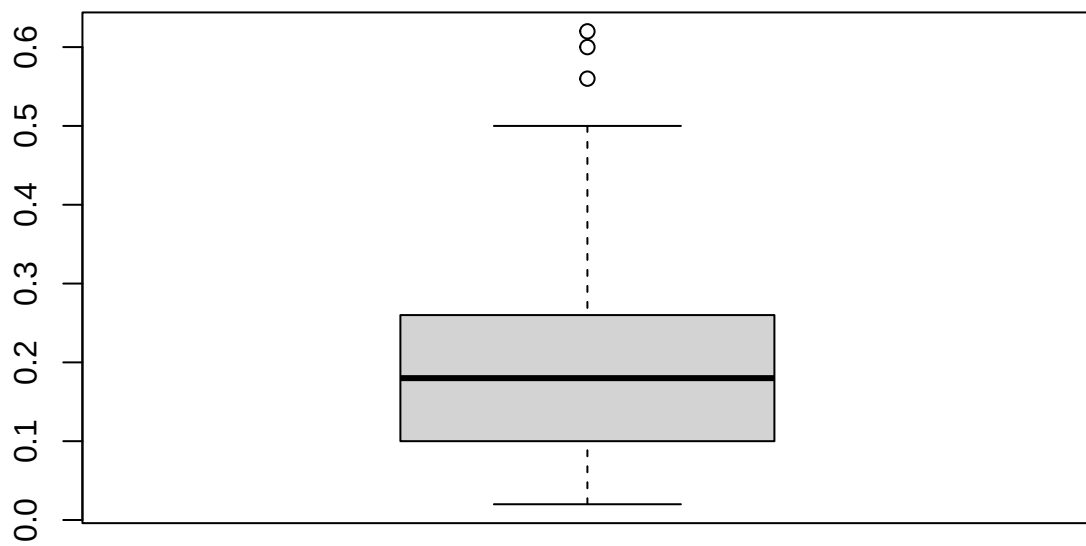


Carb.Temp

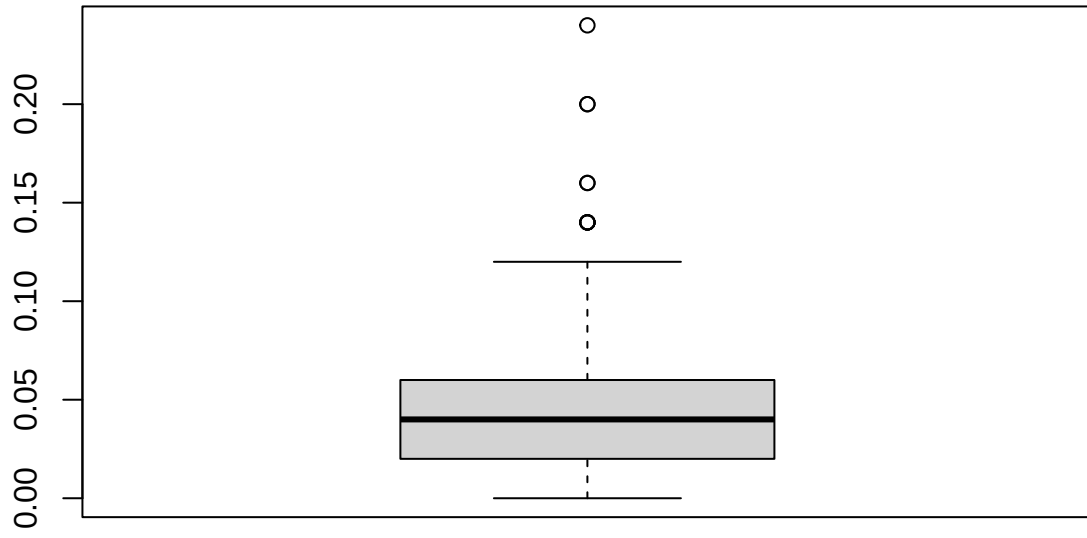




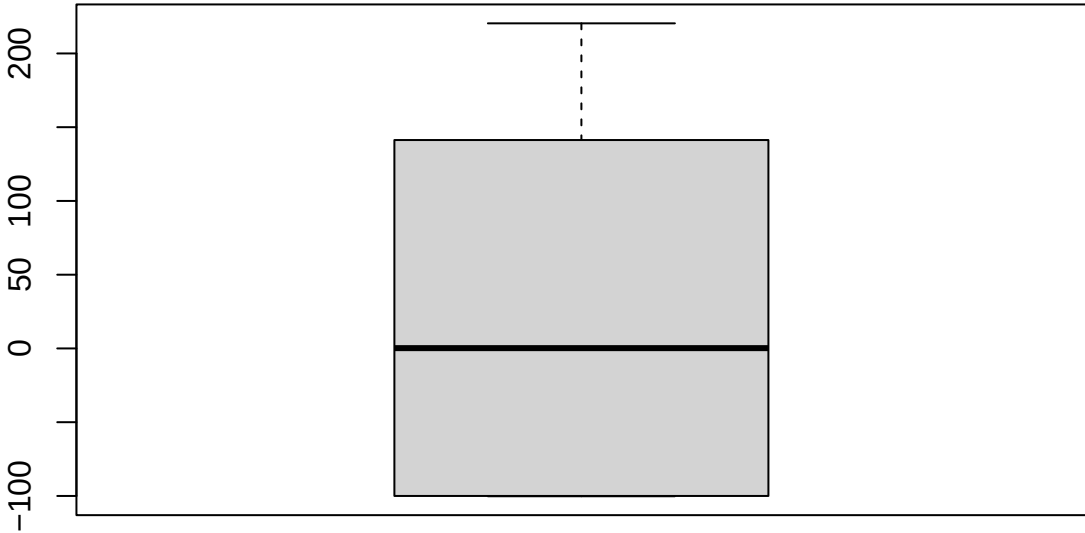
PSC.Fill



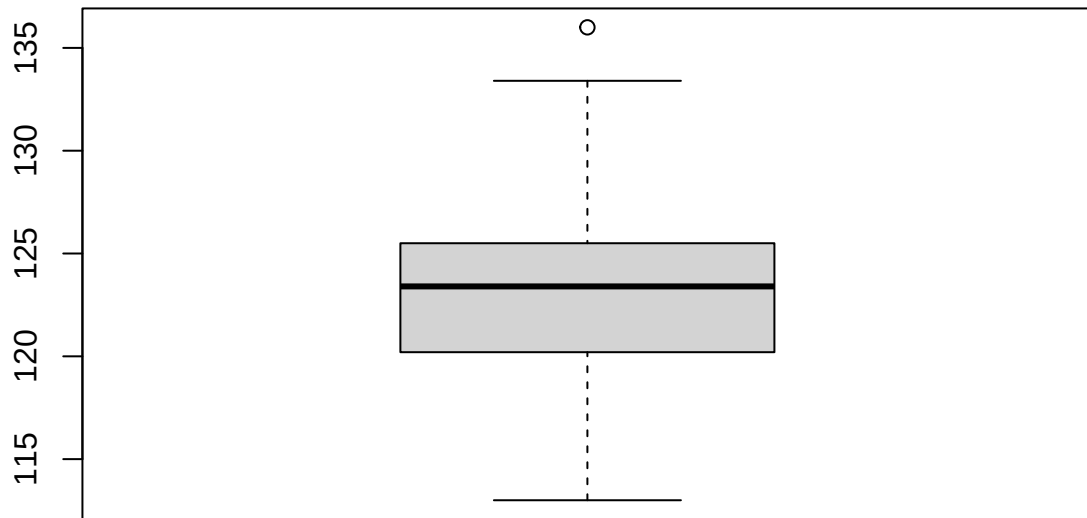
PSC.CO2



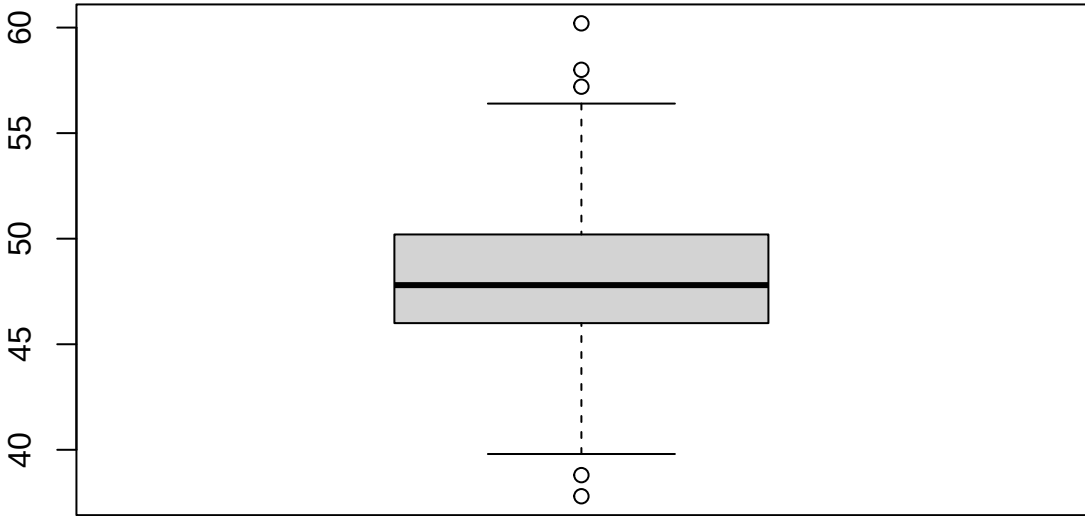
Mnf.Flow



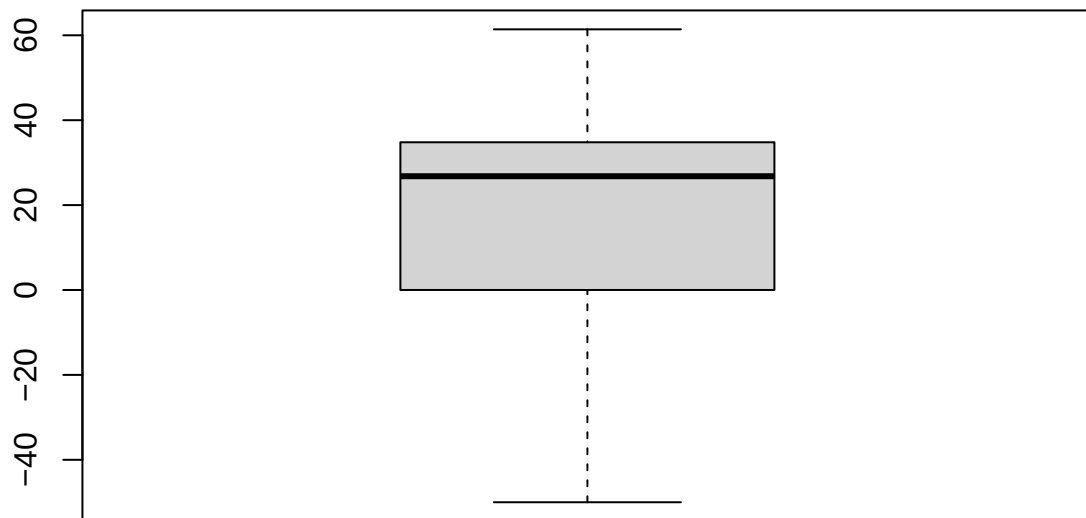
Carb.Pressure1



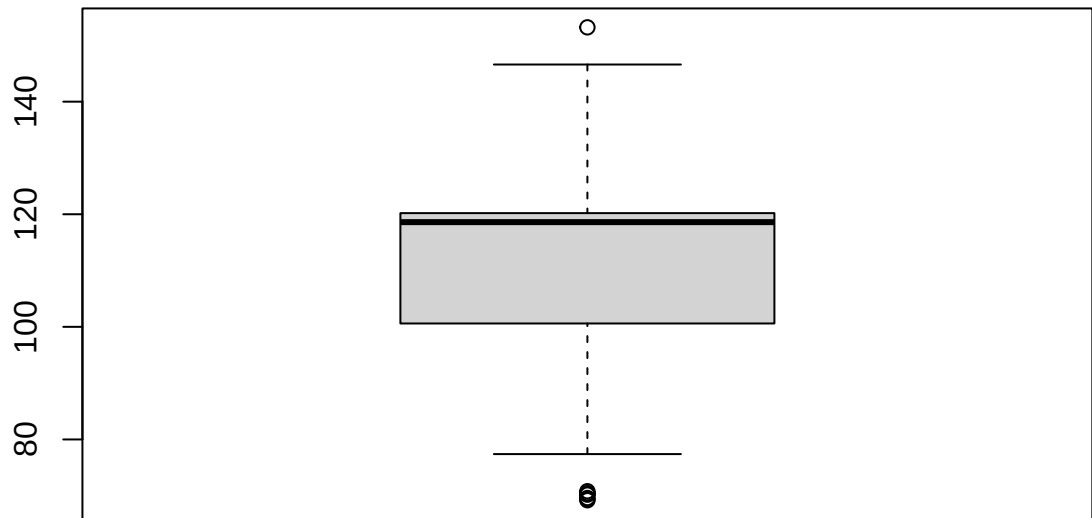
Fill.Pressure

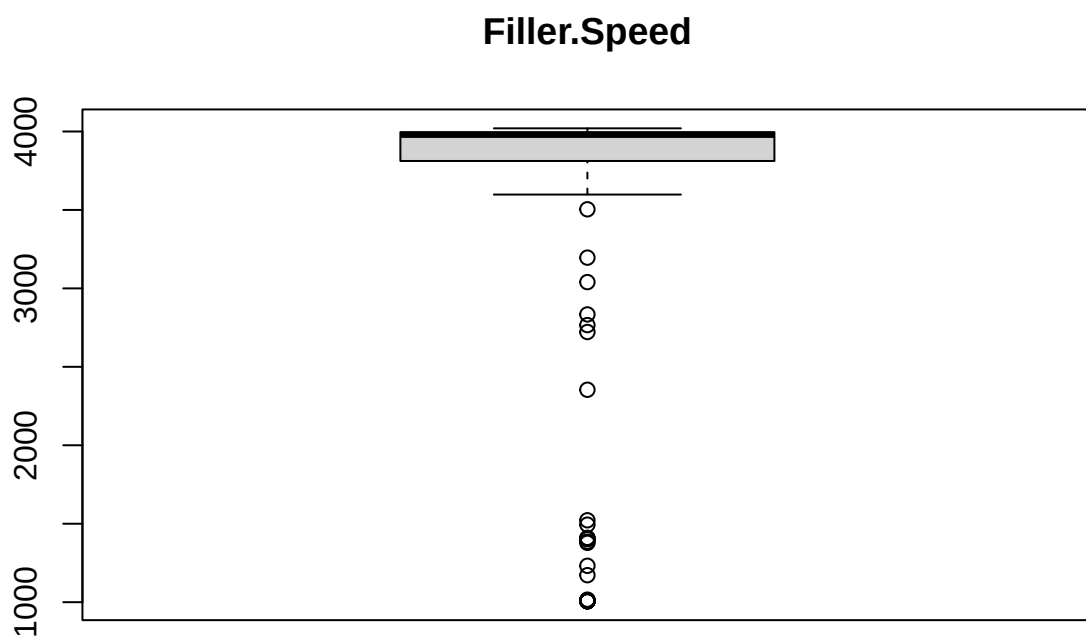


Hyd.Pressure2

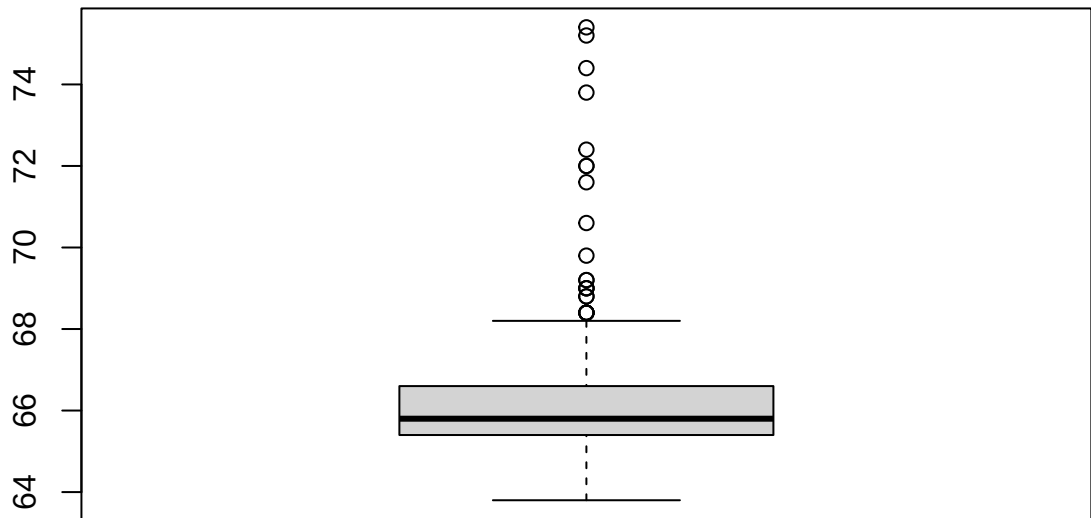


Filler.Level

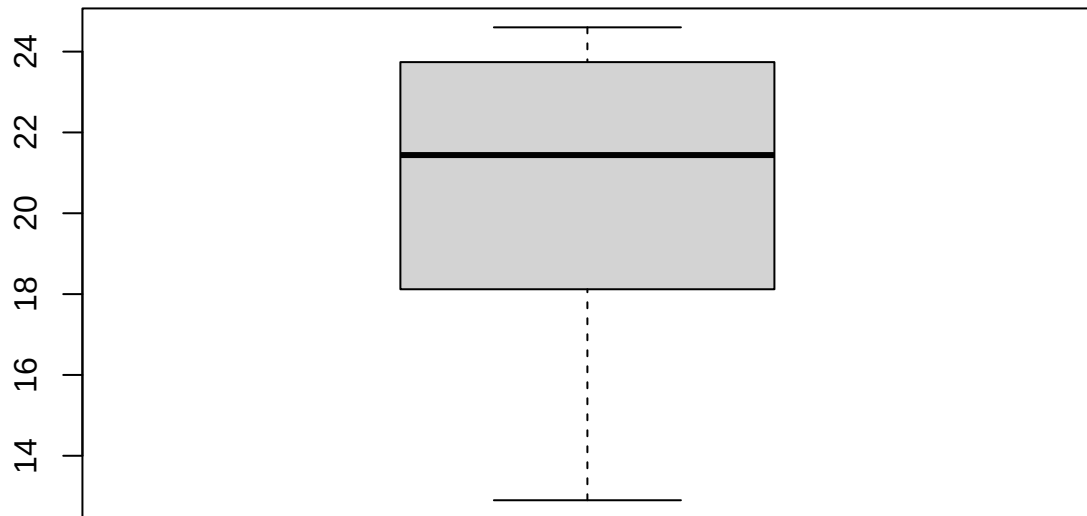


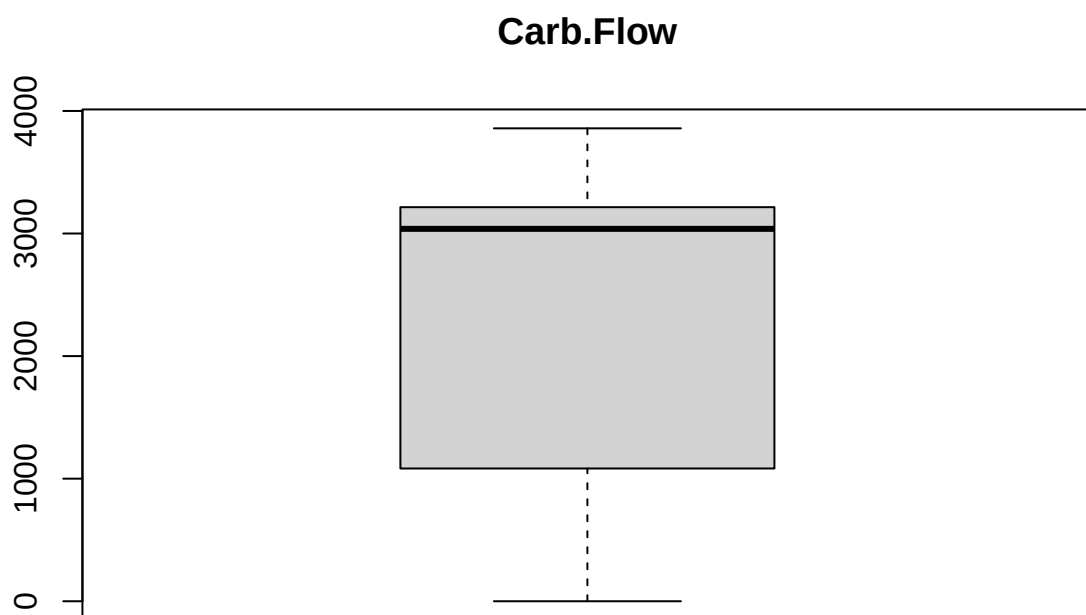


Temperature

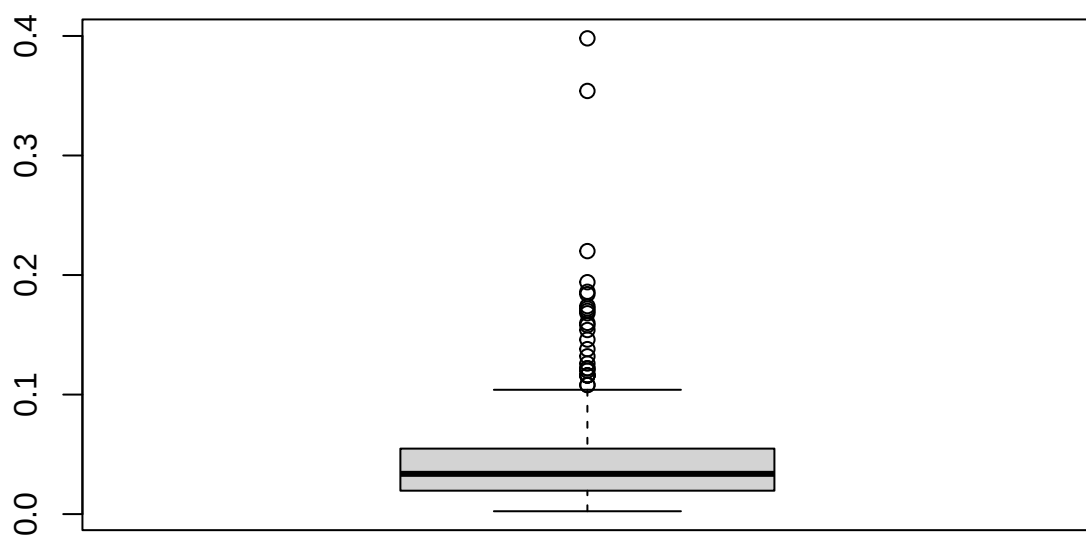


Usage.cont

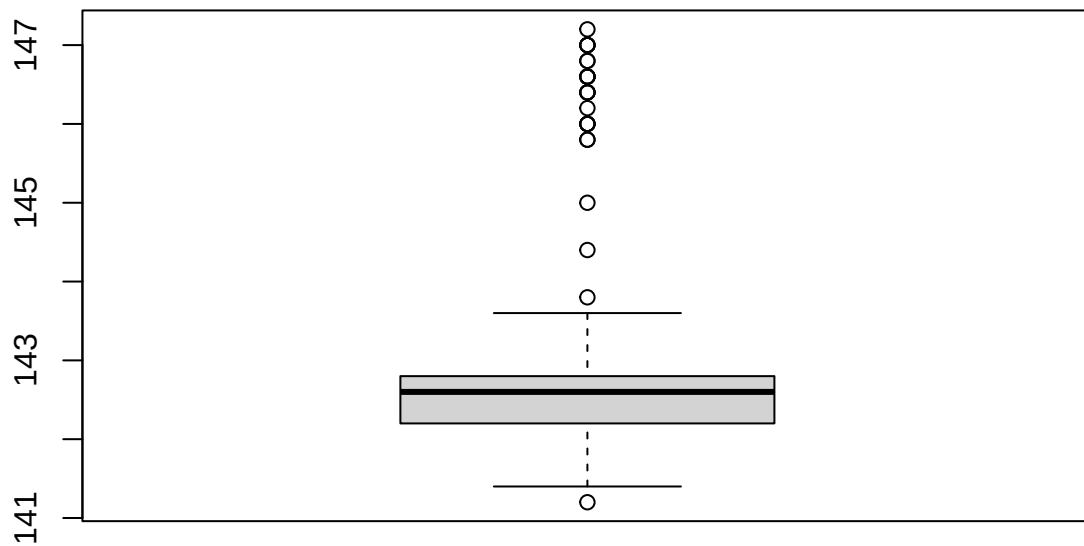




Oxygen.Filler



Air.Pressurer



```
test_data_new |> count(Brand.Code)
```

```
##   Brand.Code   n
## 1         A   35
## 2         B  129
## 3         C   31
## 4         D   64
## 5        <NA>    8
```

Oxygen filler shows up again with a few outliers, which just seems to be the case for this variable; the outliers in filler speed are all clustered; filler level has outliers on both ends, as does fill pressure. Carb.pressure1 has one high value, PSC.CO2 has a few higher values designated outliers due to the clustering on the lower end, same is true for PSC.Fill, and the rest are similar.

We won't remove any of these because they don't look like errors, and in some cases represent a different, smaller cluster of values.

Dealing with missing values

```
sum(is.na(training_data_no_na))
```

```
## [1] 515
```



```
sort(colSums(is.na(training_data_no_na)), decreasing = TRUE)
```

```
##      Brand.Code  Filler.Speed    PC.Volume    PSC.CO2  Fill.Ounces
##           120          52          39          39          38
##           PSC Carb.Pressure1 Carb.Pressure    Carb.Temp    PSC.Fill
##           33           32           27           26           23
##  Fill.Pressure Hyd.Pressure2  Filler.Level  Temperature  Oxygen.Filler
##           17           15           15           11           11
##    Carb.Volume    Usage.cont    Carb.Flow    Mnf.Flow    PH
##           10           5           2           0           0
##  Air.Pressurer
##           0
```

```
#checking the number of complete cases
table(rowSums(is.na(training_data_no_na)))
```

```
##
##      0      1      2      3      4      5      6
## 2169  309   64   18    2    2    1
```

516 missing values across 2566 cases and 21 variables - 516/53886 is about .9% overall missingness, which is very low. Notably, brand code is missing in 120 cases.

2169 cases out of 2566 are complete, or about 85%. If we dropped all NAs, we would be dropping 15% of the data. Instead, we will use the VIM package for simple KNN imputation. This package just fills in the nearest neighbor—it doesn't change the other variables. Vim can handle categorical variables, like brand code. We'll use K=5.

```
training_data_imputed <- kNN(training_data_no_na, k = 5)
```

```
#making sure it worked
sum(is.na(training_data_imputed))
```

```
## [1] 0
```

Imputing values for the test data First, let's check how many missing values there are after we've removed columns in the test data.

```
table(rowSums(is.na(test_data_new)))
```

```
##
##      1      2      3      4      5      9
## 225  33    6    1    1    1
```

```
colSums(is.na(test_data_new))
```

```
##      Brand.Code  Carb.Volume  Fill.Ounces  PC.Volume  Carb.Pressure
##           8           1           6           4           0
##    Carb.Temp    PSC    PSC.Fill    PSC.CO2    Mnf.Flow
##           1           5           3           5           0
```

```
## Carb.Pressure1  Fill.Pressure  Hyd.Pressure2  Filler.Level  Filler.Speed
##           4           2           1           2           10
##   Temperature      Usage.cont      Carb.Flow      PH  Oxygen.Filler
##           2           2           0           267           3
##   Air.Pressurer
##           1
```

```
#total NAs (subtracting 267 for the missing pH values)
sum(is.na(test_data_new)) - 267
```

```
## [1] 60
```

Now there are only 60 missing values total. 42 rows have missing values. One row has 8 missing values (the one that had 14 before). Let's remove that:

```
test_data_new <- test_data_new[rowSums(is.na(test_data_new)) != 9, ]

#confirming that only one value was removed
count(test_data_new)
```

```
##      n
## 1 266
```

Imputing with the same method we used for the training data:

```
#list all columns except PH
cols_to_impute <- setdiff(colnames(test_data_new), "PH")

test_data_imputed <- kNN(test_data_new, variable = cols_to_impute, k = 5)

#we should get 266 cases
sum(is.na(test_data_imputed))
```

```
## [1] 266
```

Break into training and test data

Breaking the training data set into an 80%/20% training and test set:

```
set.seed(1122)

train_index <- sample(nrow(training_data_imputed),
                     0.8 * nrow(training_data_imputed))

#create a training set
train_y <- training_data_imputed$PH[train_index]
train_x <- training_data_imputed[train_index, ] |> dplyr::select(-PH)

#test set
test_y <- training_data_imputed$PH[-train_index]
test_x <- training_data_imputed[-train_index, ] |> dplyr::select(-PH)
```

Centering and scaling

I did not scale anything individually, every package should have an option to center/scale automatically, and use a Box-Cox for the data with less normal distributions.